

CHM 3120: Introduction to Analytical Chemistry

Unit 2 - Section 1.1:

Precision and Accuracy of Chemical Analyses

Chapters 3-4

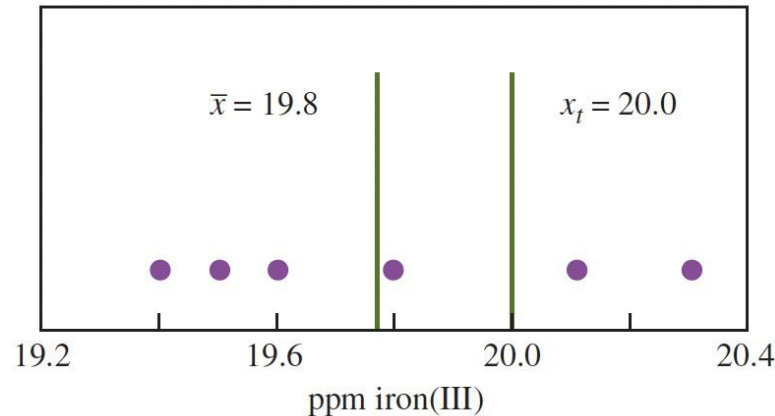
SPRING 2026

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Error

There is error associated with every measurement. The best we can do is to carefully apply reliable techniques to get as close to the “true” value as possible.



Six replicate measurements were made on a 20.0 ppm iron(III) standard. The average result was 19.78 ppm, rounded to 19.8 ppm (based on significant figures, to be discussed later).

Repetition of a method can get us measurements that are **precise** (*i.e.*, have good reproducibility).

Agreement of measurements made by different methods gives us **accuracy**.

Really, we want measurements that are fit for purpose...

Fit for Purpose

Do I have cancer?

Am I pregnant?

Is the sample positive for cocaine?

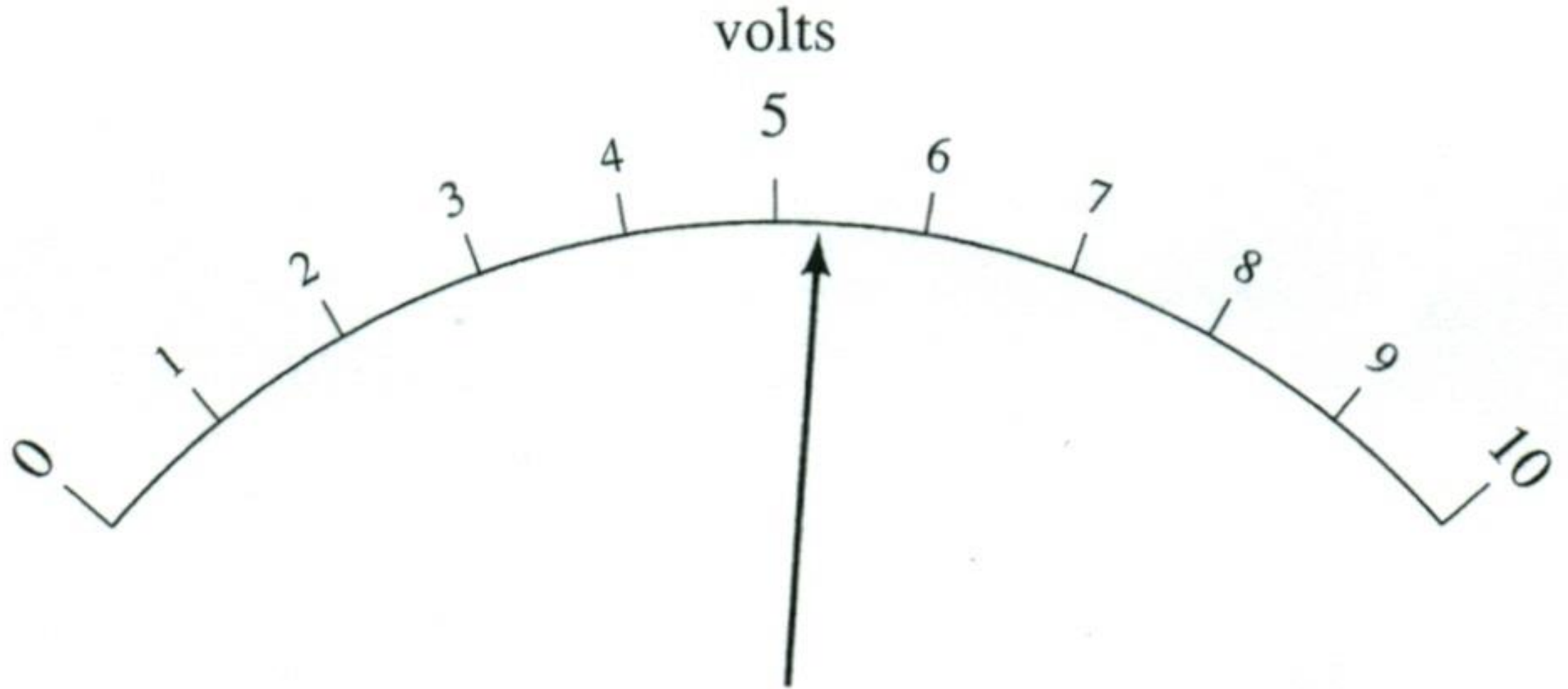
Is the sample positive for marijuana?

Is the sample positive for steroids?

Estimating the reliability of experimental data is essential: data without known quality are worthless.

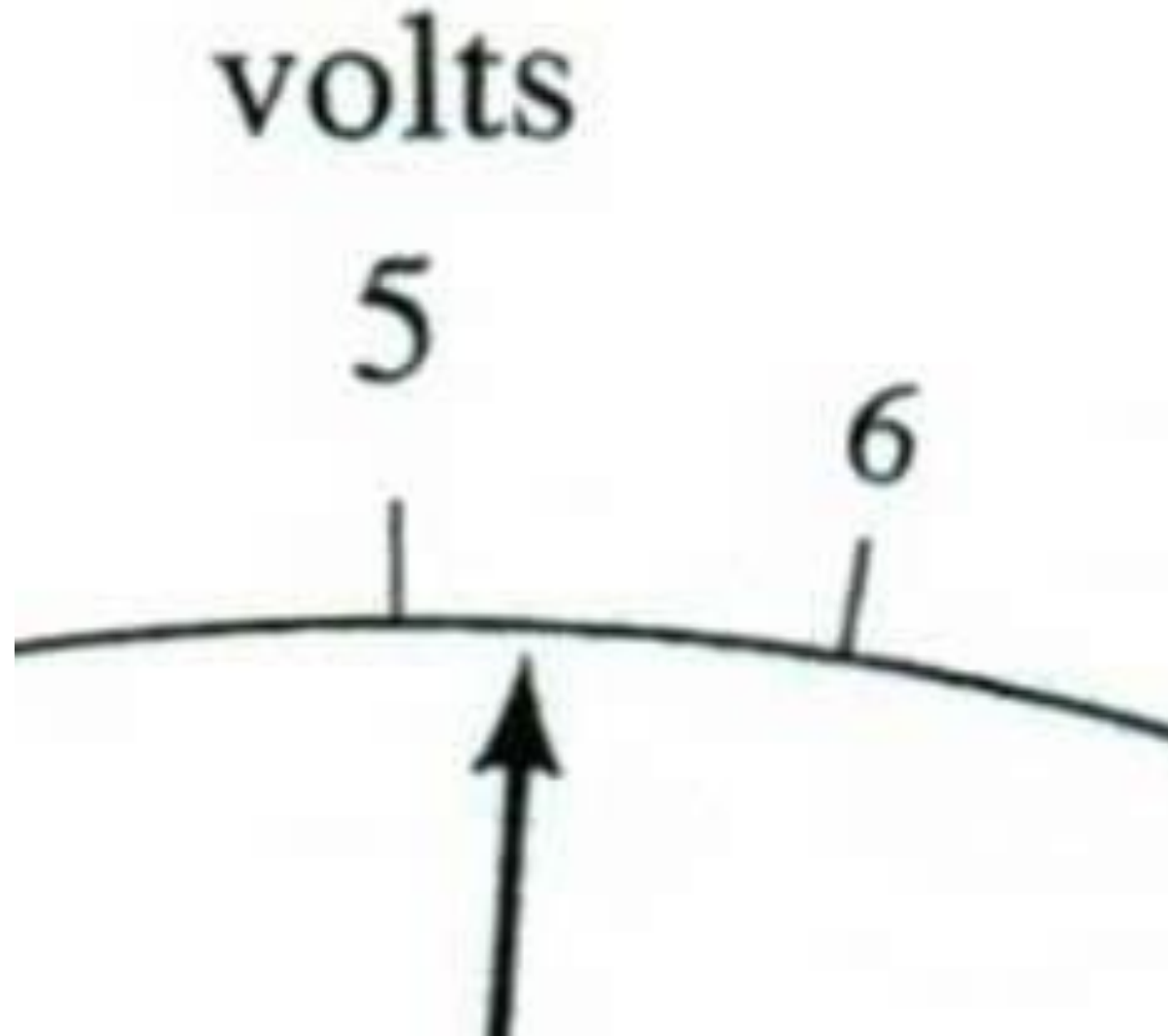
But even less accurate results can be valuable if the **limits of uncertainty** are known.

Reading a Scale

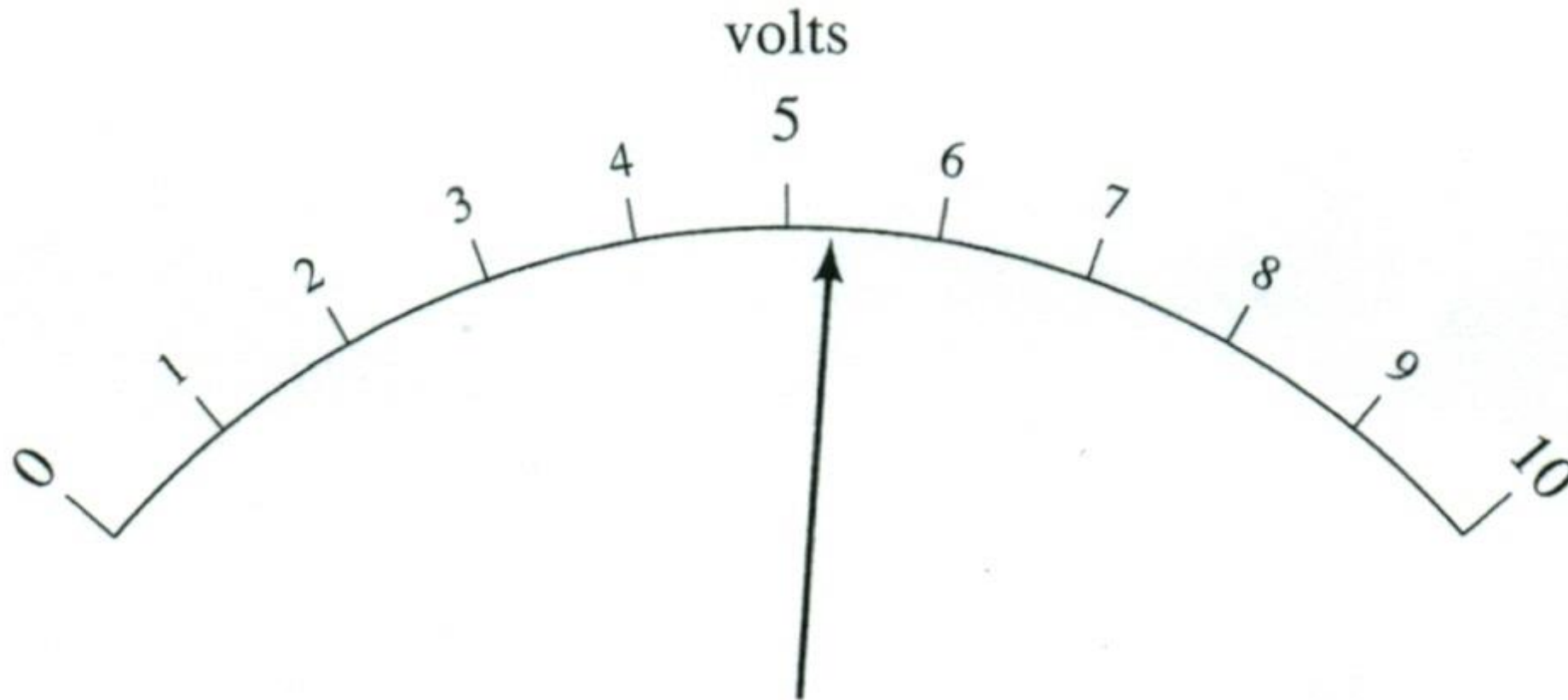


What would you estimate this reading to be?

Reading a Scale



Reading a Scale



Best estimate of the voltage = 5.3 volts

Probable range = 5.2 – 5.4 volts

What maximum error can be tolerated in the result?

The Mean and the Median

- To improve reliability and assess variability, 2–5 replicates of a sample are typically analyzed.
- Since individual results vary, we use a central value (the mean or median) as the “best” **estimate**.
- Analyzing replicates justifies the extra effort because the central value is more reliable, and it helps us estimate the **limits of uncertainty**.

Mean: The average of the total measurements

The sum of the measurements divided by the total number of measurements:

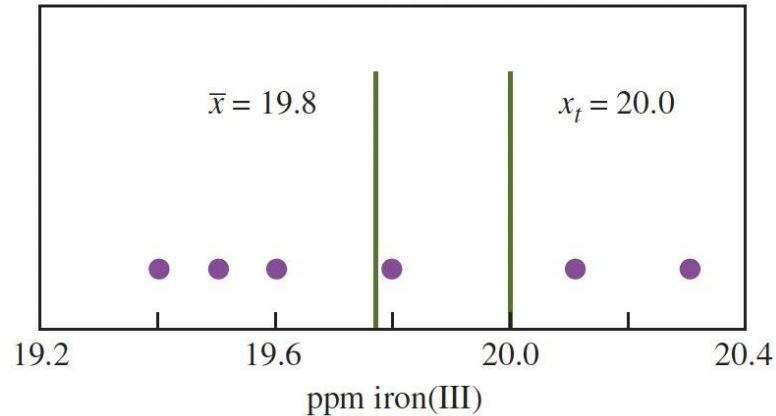
$$\bar{x} = \frac{\sum x_i}{n}$$

Median: The median is the middle result when replicate data are arranged in **increasing or decreasing** order.

Sum: represented by the Greek letter sigma (Σ)

Mean: represented by x bar ($\bar{x}$).

The Mean and the Median: Sample Problem



$$\text{mean} = \bar{x} = \frac{19.4 + 19.5 + 19.6 + 19.8 + 20.1 + 20.3}{6} = 19.78 \approx 19.8$$

Because the set contains an even number of measurements, the median is the average of the central pair:

$$\text{median} = \frac{19.6 + 19.8}{2} = 19.7$$

The median is used when a **set of data contain an outlier**, a result that differs significantly from others in the set. An outlier can have a significant effect on the mean of the set but has no effect on the median.

Precision

Precision is the closeness of results **to others** obtained in exactly the same way: how much an individual result (x_i) differs from the mean $\bar{x}$ (deviation from the mean d_i)

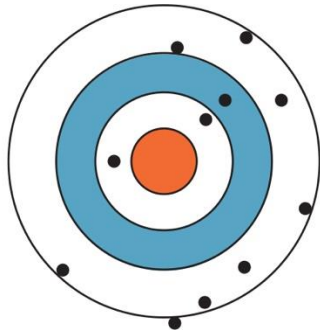
$$d_i = |x_i - \bar{x}|$$

Precision can be determined by measuring **replicate** samples.

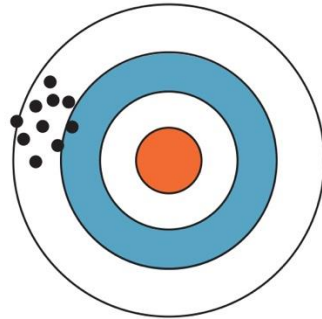
Terms used to describe the precision of a set of replicate data include **standard deviation**, **variance**, and **coefficient of variation** (will be discussed shortly).

Accuracy

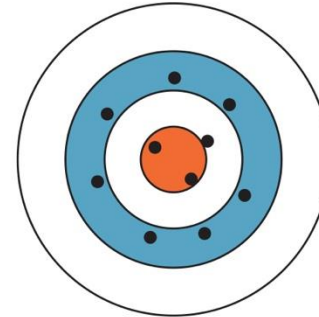
Accuracy is the closeness of a measured value **to the true or accepted value** and is expressed by the error.



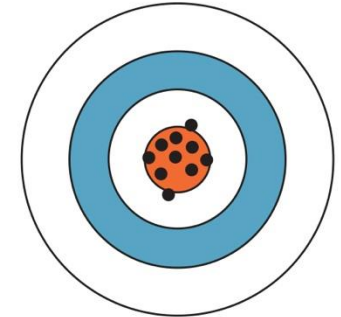
Low accuracy, low precision



Low accuracy, high precision



High accuracy, low precision



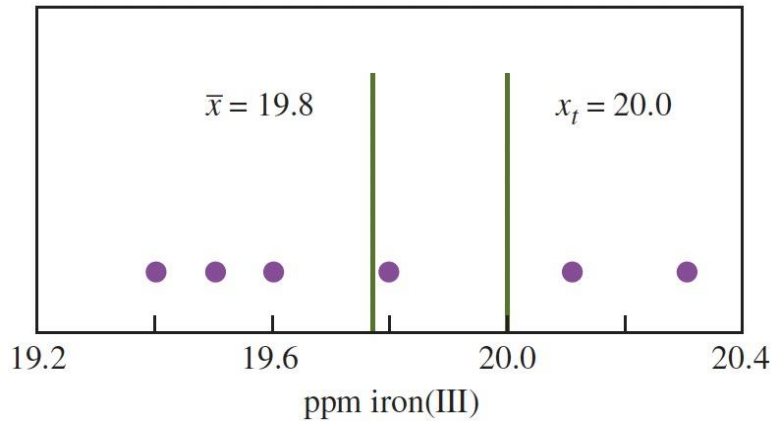
High accuracy, high precision

Accuracy is more difficult to determine because the true value is usually unknown. An accepted value must be used instead.

Accuracy

Absolute Error

$$E = x_i - x_t$$



$$19.4 \rightarrow E = 19.4 - 20.0 = -0.6$$

$$19.8 \rightarrow E = 19.8 - 20.0 = -0.2$$

$$20.3 \rightarrow E = 20.3 - 20.0 = 0.3$$

Relative Error

$$E_r = \frac{x_i - x_t}{x_t} \times 100\%$$

Relative error is also expressed in %, parts per thousand (ppt), or parts per million (ppm)

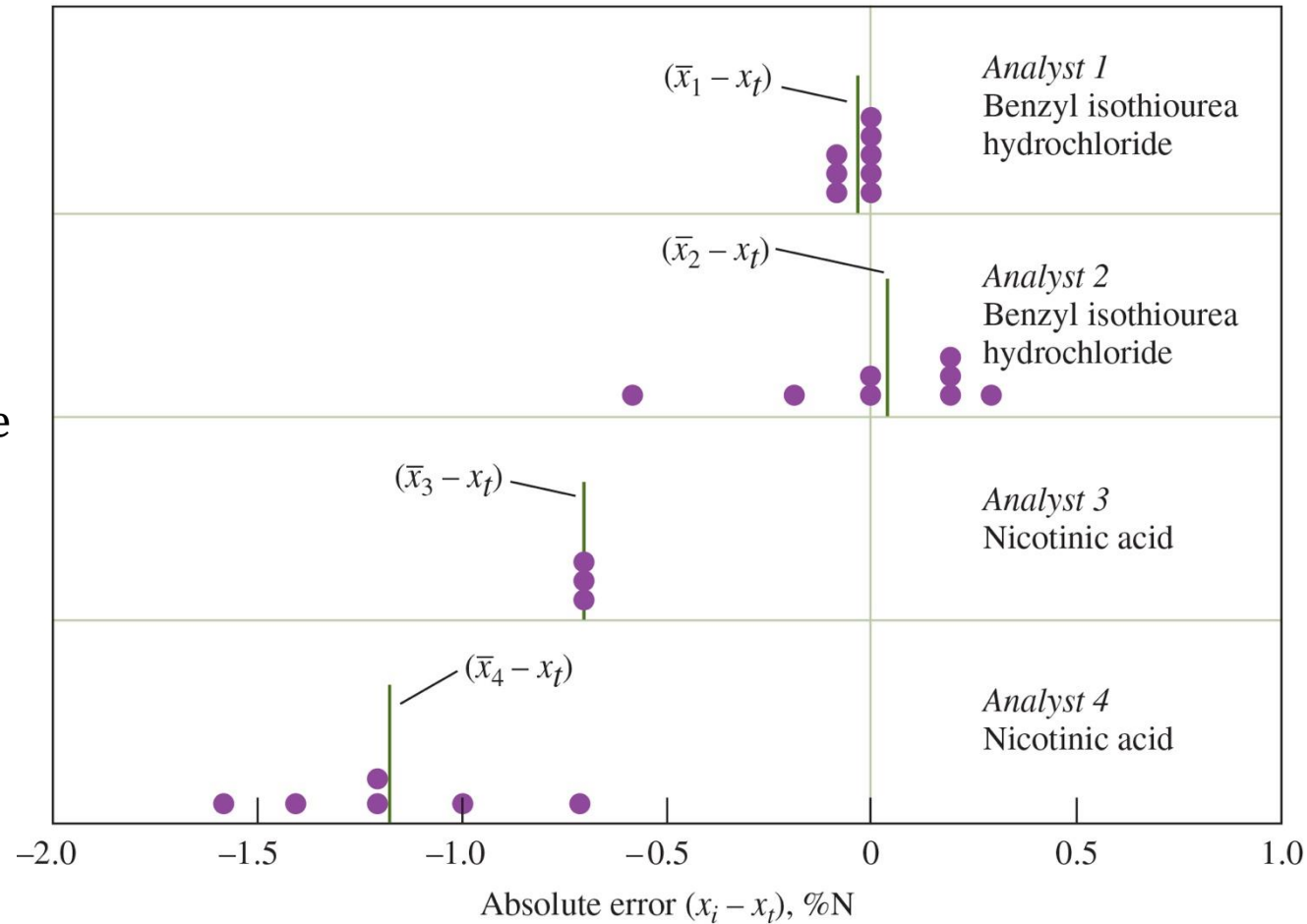
Ex:

$$E_r = \frac{19.8 - 20.0}{20.0} \times 100\% = -1\% = -10 \text{ ppt}$$

$$E_r = \frac{20.3 - 20.0}{20.0} \times 100\% = 1.5\% = 15 \text{ ppt}$$

Types of Errors in Experimental Data

Determination of nitrogen in two compounds.
The dots show the absolute errors of replicate results obtained by four analysts



High precision and high accuracy

Poor precision but good accuracy

Excellent precision, but significant error in the numerical average for the data

Poor precision and poor accuracy

Precise results are not always accurate

Types of Errors in Experimental Data

- **Random (Indeterminate) Error:** Causes results to scatter around a mean, often symmetrically. It affects the precision.
 - Analysts 1 and 3 have lower random error than analysts 2 and 4.
- **Systematic (Determinate) Error:** Shifts the average result away from the true value, making all results too high or too low. It affects accuracy.
 - Analysts 3 and 4 show systematic errors ($\sim 20.7\%$ and 21.2% nitrogen), while analysts 1 and 2 show minimal systematic error.
- **Gross Errors:** Occur occasionally, are often large, and result from human mistakes (e.g., misreading instruments or calculation errors). They lead to outliers—data points that differ significantly from the rest.

Systematic Error

Reproducible errors resulting from experimental flaws (Instrumental, Method, or Personal errors)

Generally detectable and correctable

There is no simple way to predict systematic error

Systematic errors lead to bias.

- Bias measures the systematic error associated with an analysis.
 - Bias has a negative sign if it causes results to be low.
 - Bias has a positive sign if it causes results to be high.



Random: small
Systematic: ?

(a)



Random: small
Systematic: ?

(b)



Random: large
Systematic: ?

(c)



Random: large
Systematic: ?

(d)

Systematic Error

How to discover systematic errors?

Use a **Known Standard**: Test the procedure with a sample of known composition (e.g., standard mass).

Run a **Blank**: Analyze a blank sample (should give zero); any signal indicates contamination or bias.

Use a **Different Method**: Cross-check using a different technique or instrument (e.g., different balance or method).

Round Robin Test: Send identical samples to multiple labs and compare results to reveal consistent bias.

Random Error

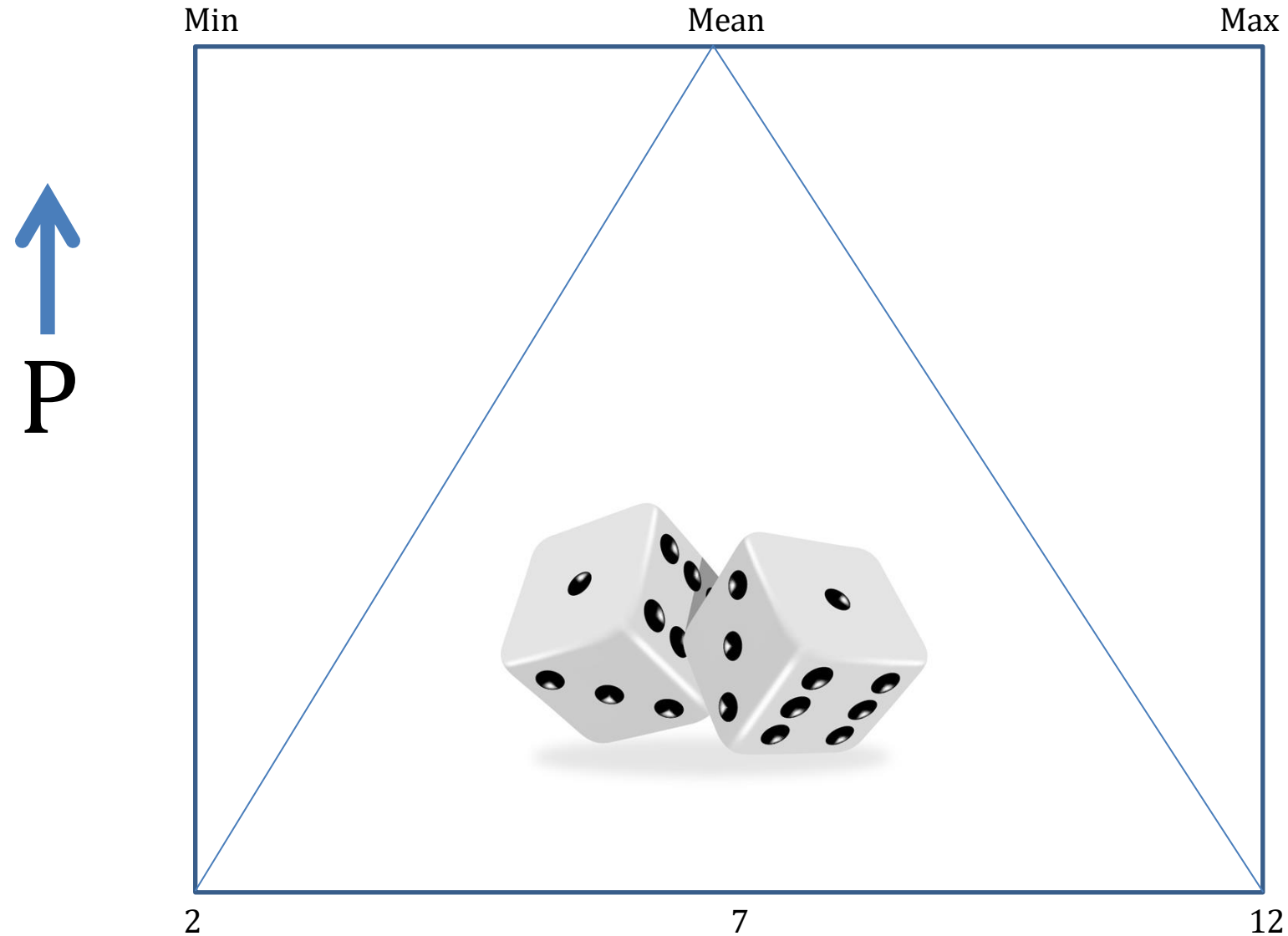
Equally **likely** to be positive or negative

Always present and cannot be corrected

Will tend to decrease **with increasing** observations

Rather **easy to quantify** and study with statistical tools

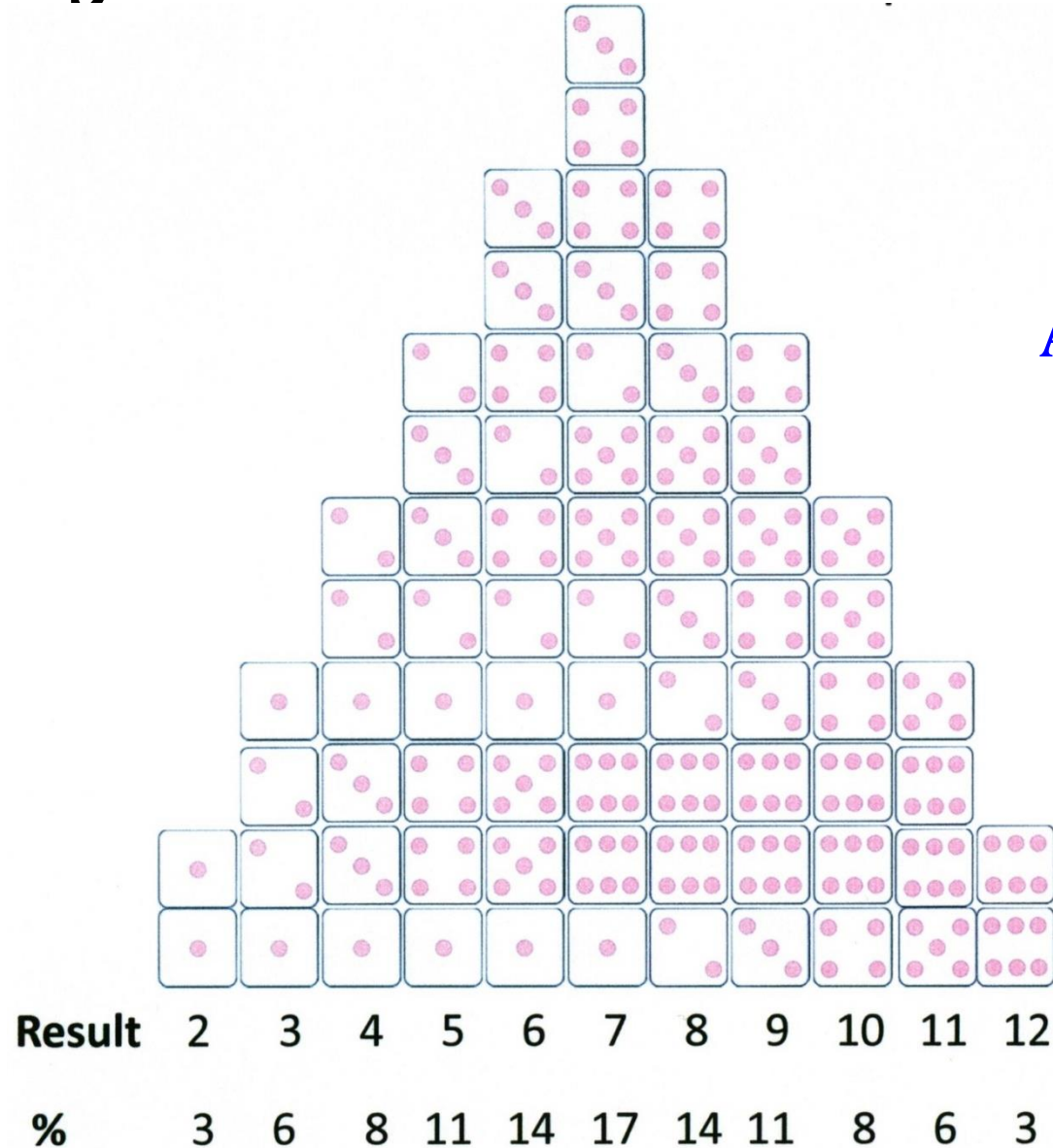
Probability of A Random Event



Probability of A Random Event

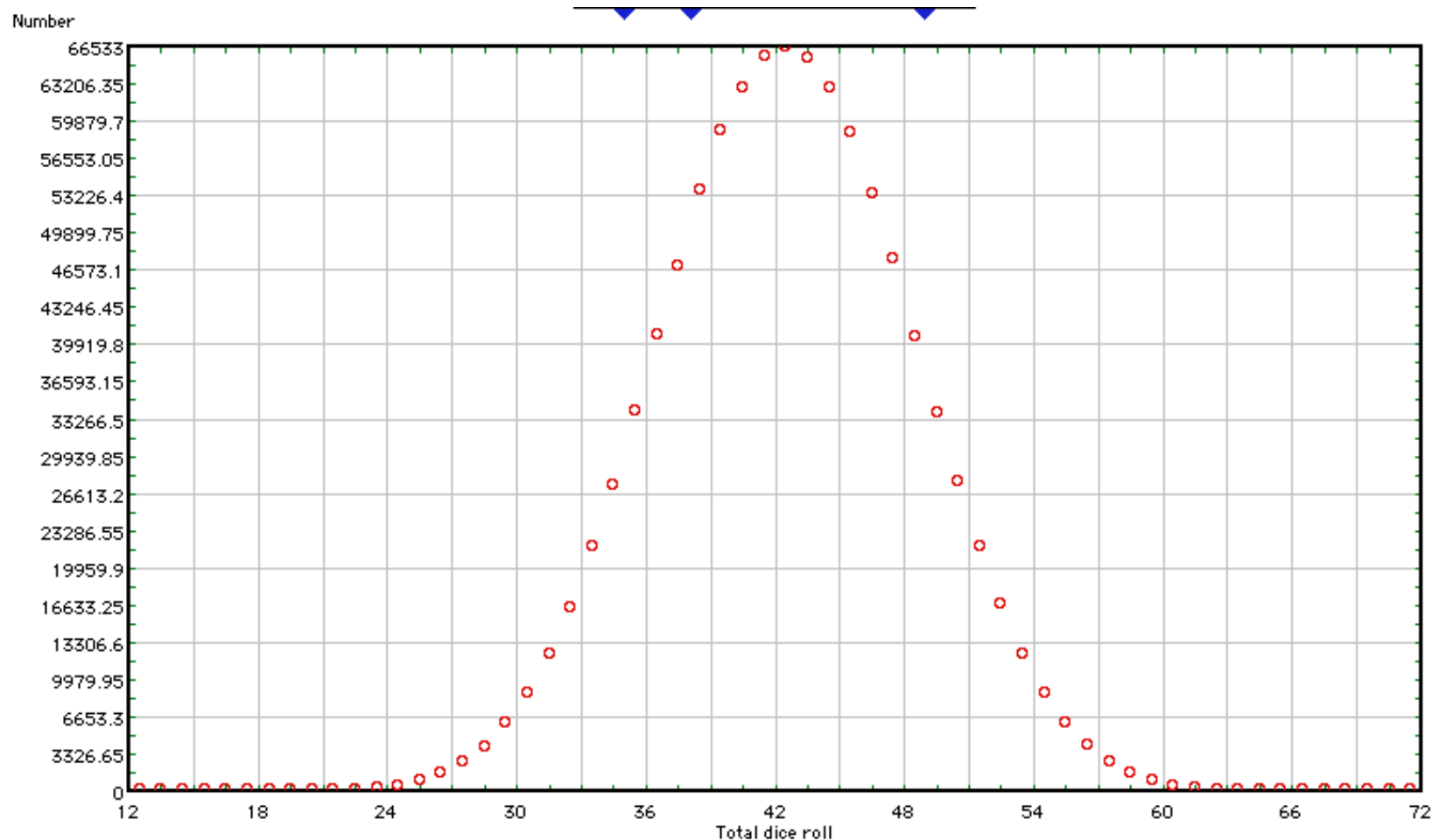
All probabilities for the
tossing of two dice

A primitive binomial
distribution!



Probability of A Random Event

Results for 1,000,000 Rolls of 12 dice

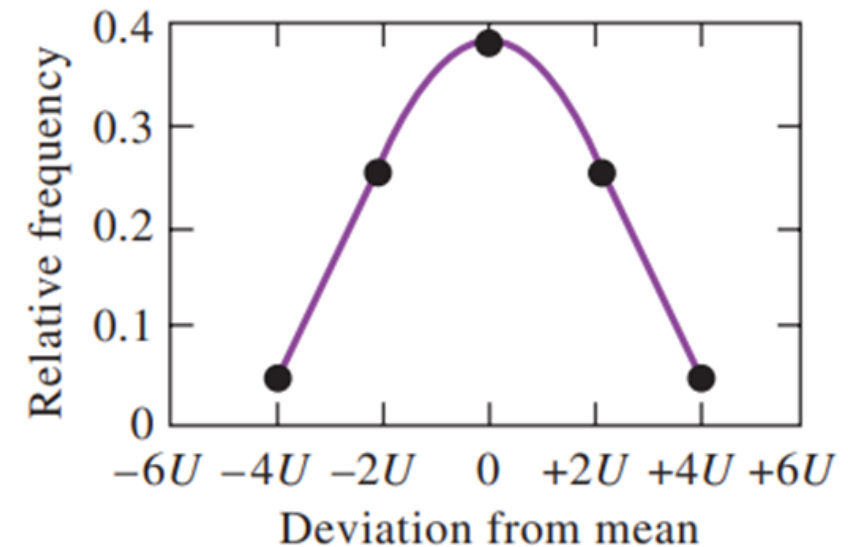


Random Error

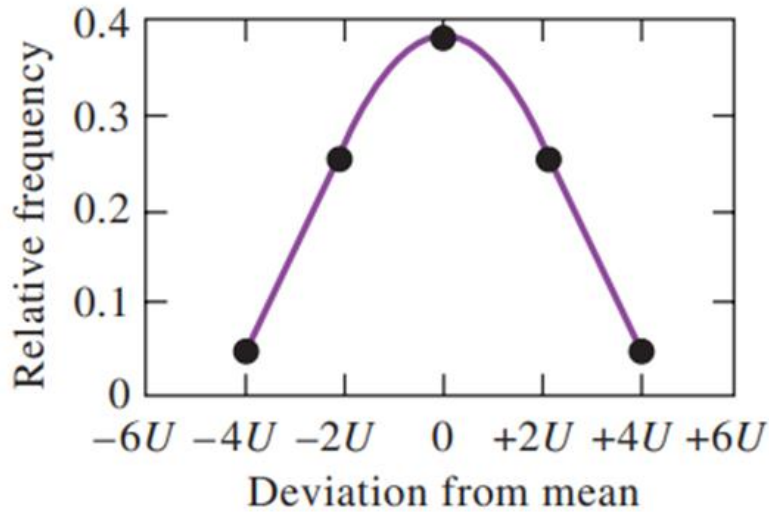
Even small, undetectable uncertainties can combine to create a detectable random error.

For example, if four small errors each have an equal **probability** of being $+U$ or $-U$, their random combinations can lead to a noticeable overall error — showing **how random errors accumulate** through probability.

Possible Combinations of Four Equal-Sized Uncertainties (U)			
Combinations of Uncertainties	Magnitude of Random Error	Number of Combinations	Relative Frequency
$+ U_1 + U_2 + U_3 + U_4$	$+ 4U$	1	$1/16 = 0.0625$
$- U_1 + U_2 + U_3 + U_4$	$+ 2U$	4	$4/16 = 0.250$
$+ U_1 - U_2 + U_3 + U_4$			
$+ U_1 + U_2 - U_3 + U_4$			
$+ U_1 + U_2 + U_3 - U_4$			
$- U_1 - U_2 + U_3 + U_4$	0	6	$6/16 = 0.375$
$+ U_1 + U_2 - U_3 - U_4$			
$+ U_1 - U_2 + U_3 - U_4$			
$- U_1 + U_2 - U_3 + U_4$			
$- U_1 + U_2 + U_3 - U_4$	$- 2U$	4	$4/16 = 0.250$
$+ U_1 - U_2 - U_3 + U_4$			
$- U_1 + U_2 - U_3 - U_4$			
$- U_1 - U_2 + U_3 - U_4$			
$- U_1 - U_2 - U_3 + U_4$	$- 4U$	1	$1/16 = 0.0625$
$- U_1 - U_2 - U_3 - U_4$			

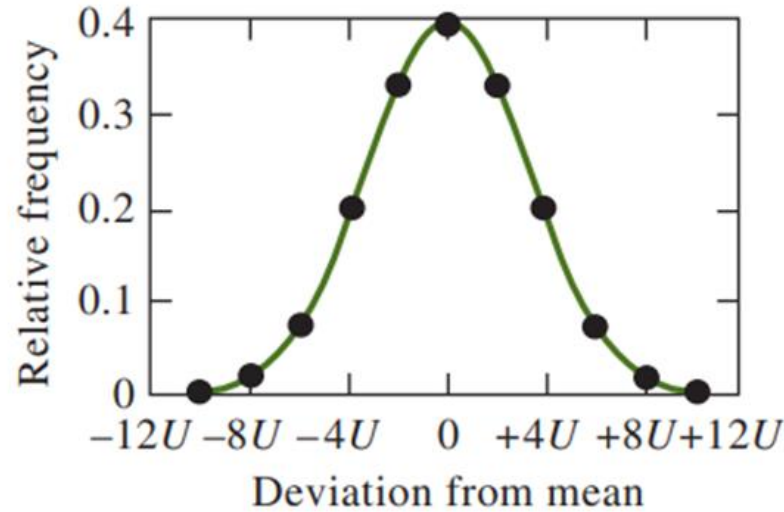


Gaussian Curve (Normal Error Curve)



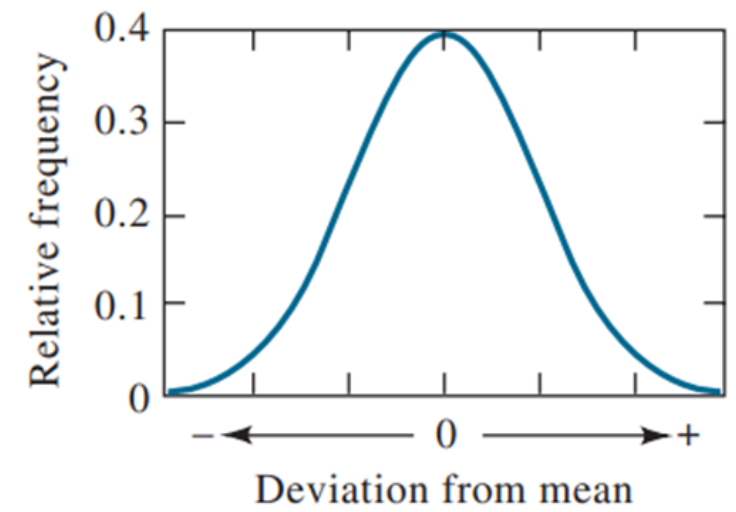
Small N

→



Larger N

→

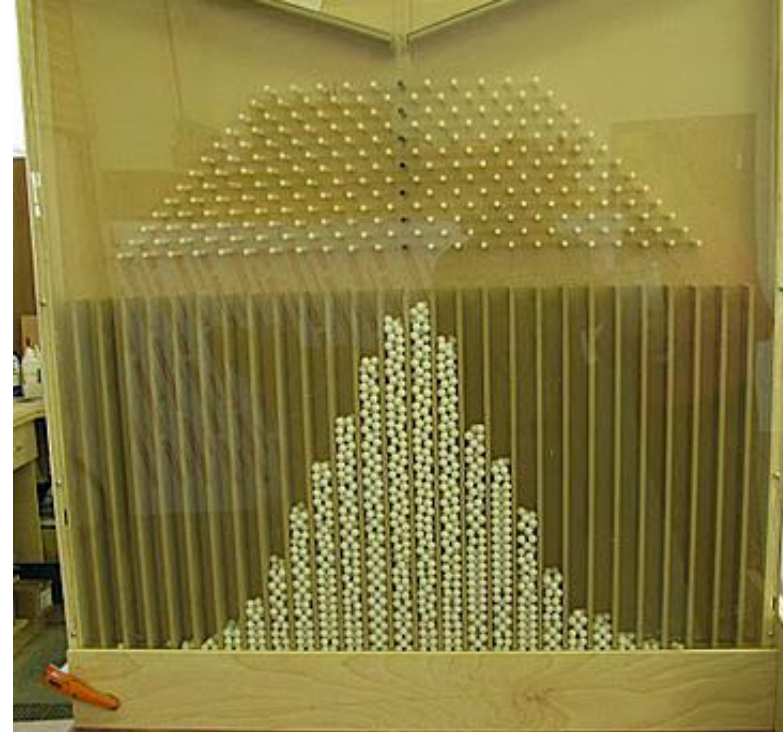
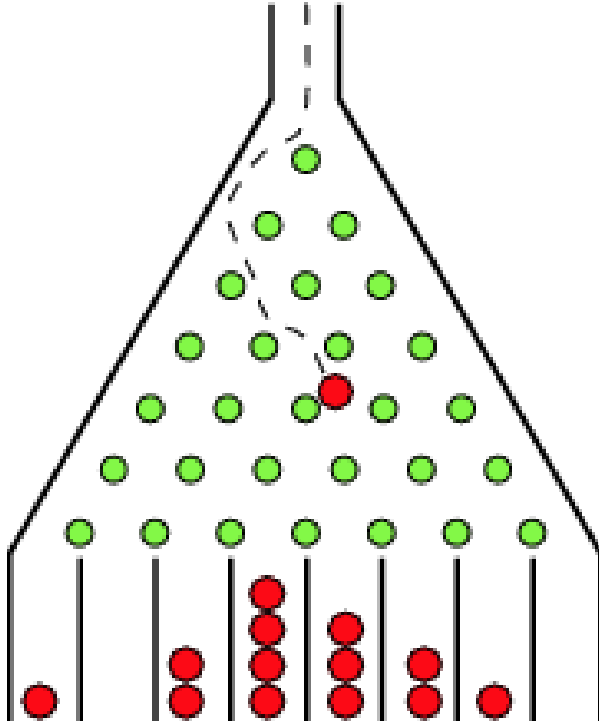


Very Large N

Increasing number of measurements (N) → small deviations occur more frequently;
large deviations become less probable

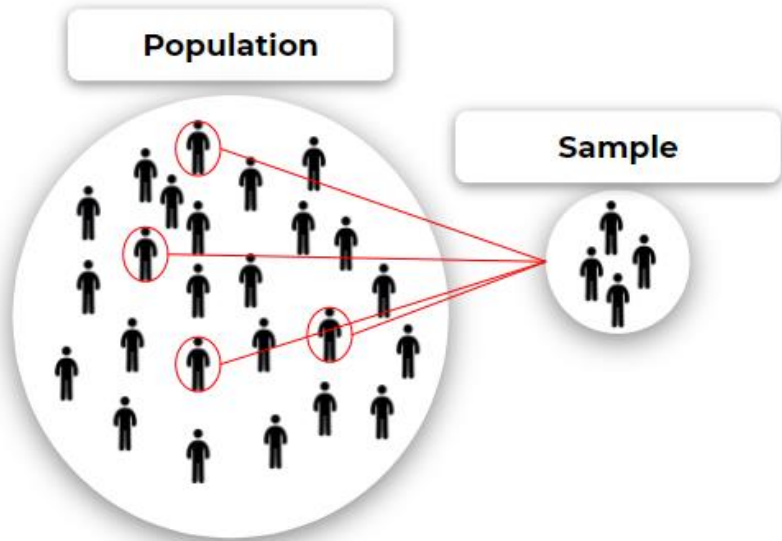
When a large number of individual measurement errors are plotted, a symmetric **bell-shaped (Gaussian)** distribution is obtained, centered at the mean.

Free Learning: “Galton” Board



[Link](#)

Samples and Populations



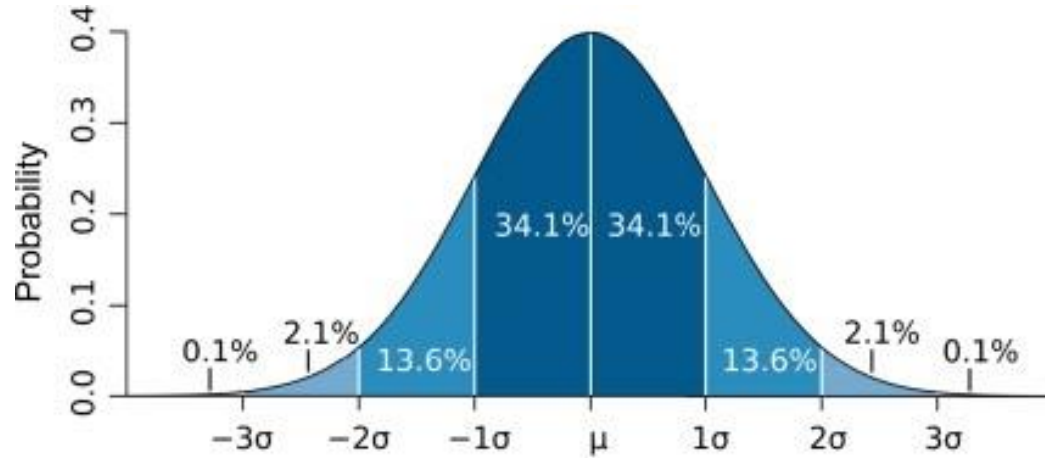
- **Populations** are defined by the experimenter and represent all measurements of interest (may be real or hypothetical).
- **Samples** are subsets of populations.

Because it is often impossible to collect data on an entire population, samples from a population are commonly used to make inferences.

Statistical laws are originally developed for entire populations, but can be applied to samples **with some modifications**. This is especially important for small samples, which may not fully represent the population.

We'll first look at Gaussian statistics for populations, then see how to adjust them for small sample sizes.

Properties of Gaussian Curves



PDF: Probability Distribution Function

$$Probability(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[-(x - \mu)^2 / (2\sigma^2)\right]}$$

A population mean (μ) is the true mean for the entire population.

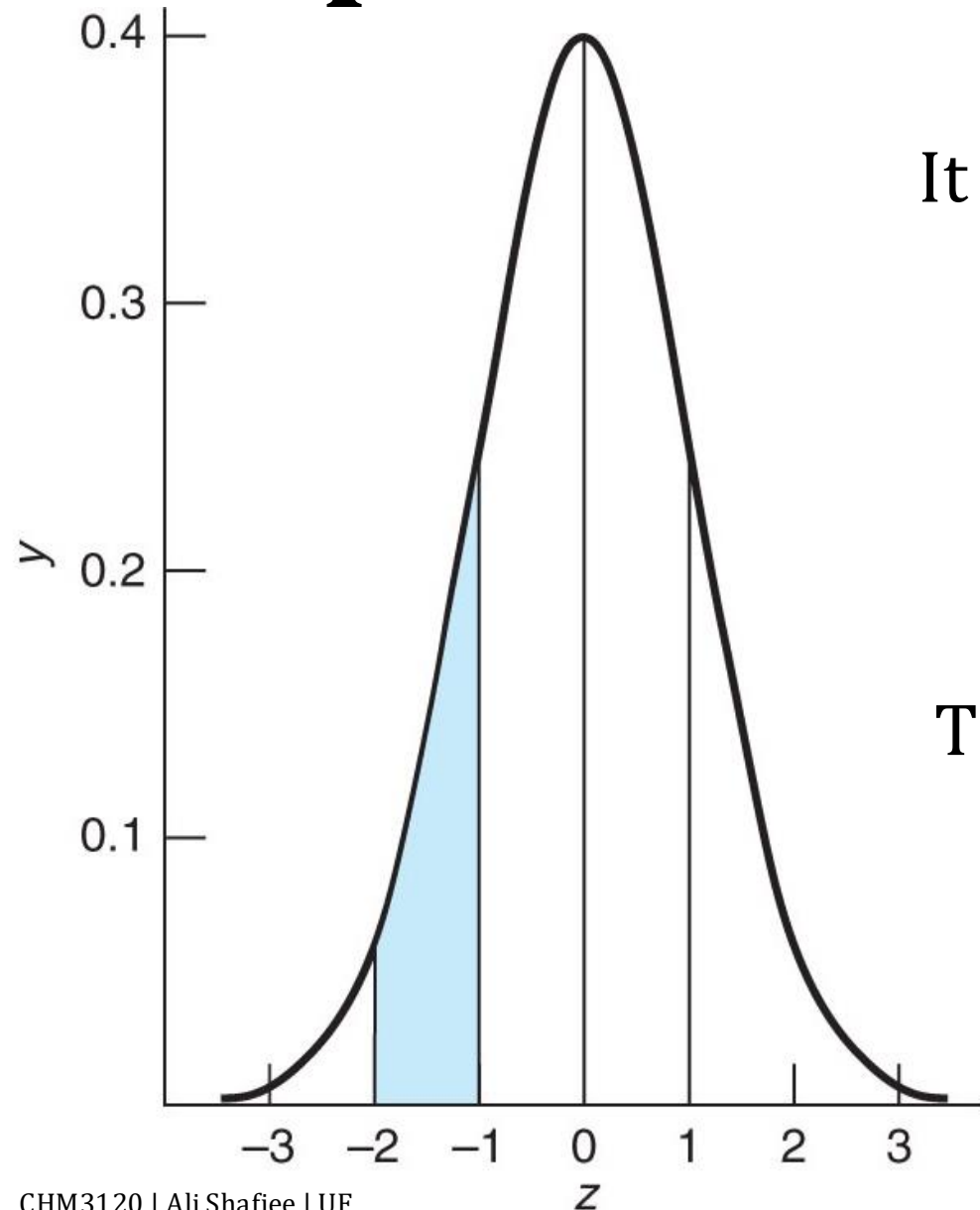
Population standard deviation (σ) is a measure of the precision of the population.

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

The square of the standard deviation is called the variance (σ^2)

N: number of data points
making up the entire population

Properties of Gaussian Curves

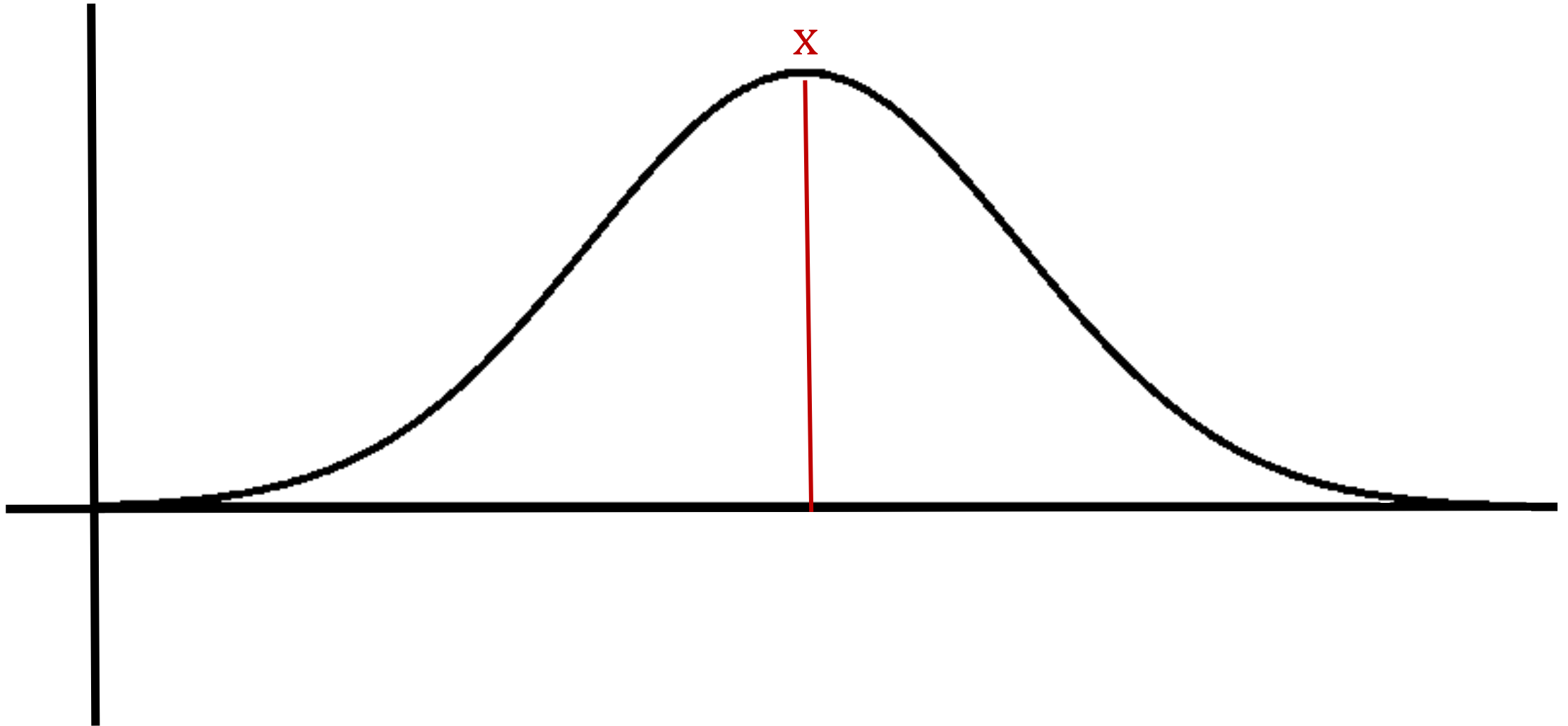


It is useful to express deviations from the mean in multiples of z , of the standard deviation:

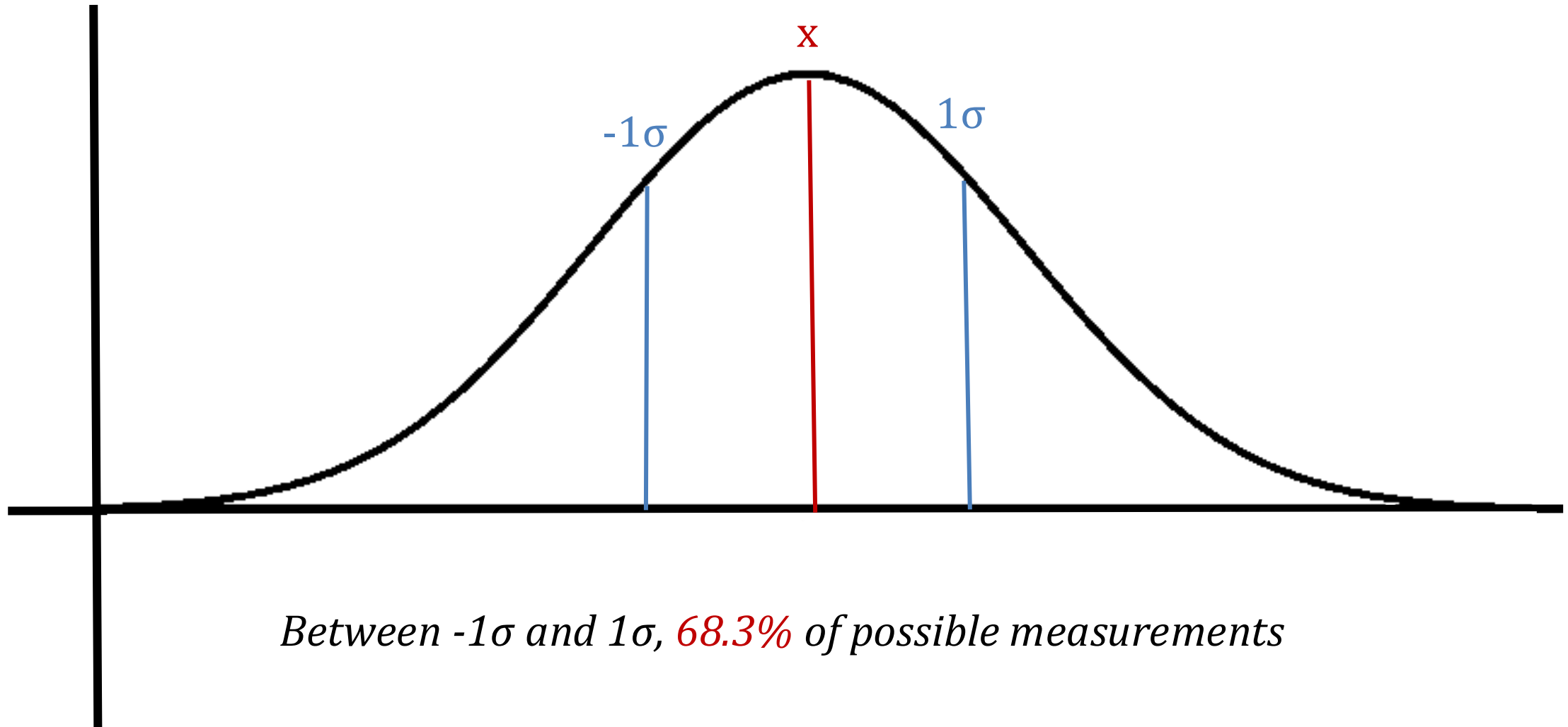
$$z = \frac{x - \mu}{\sigma}$$

The probability of measuring z in a certain range is equal to the area of that range

Standard Deviations

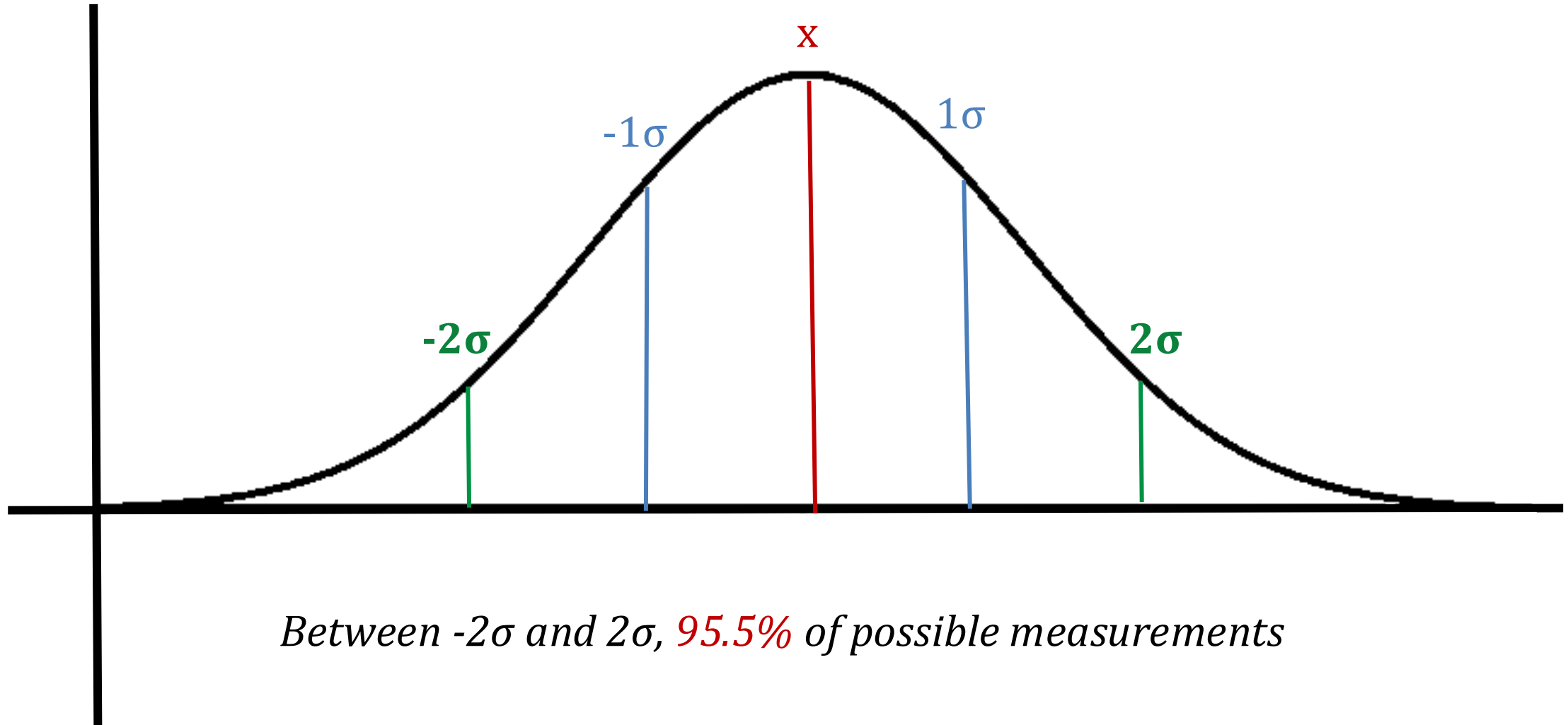


Standard Deviations

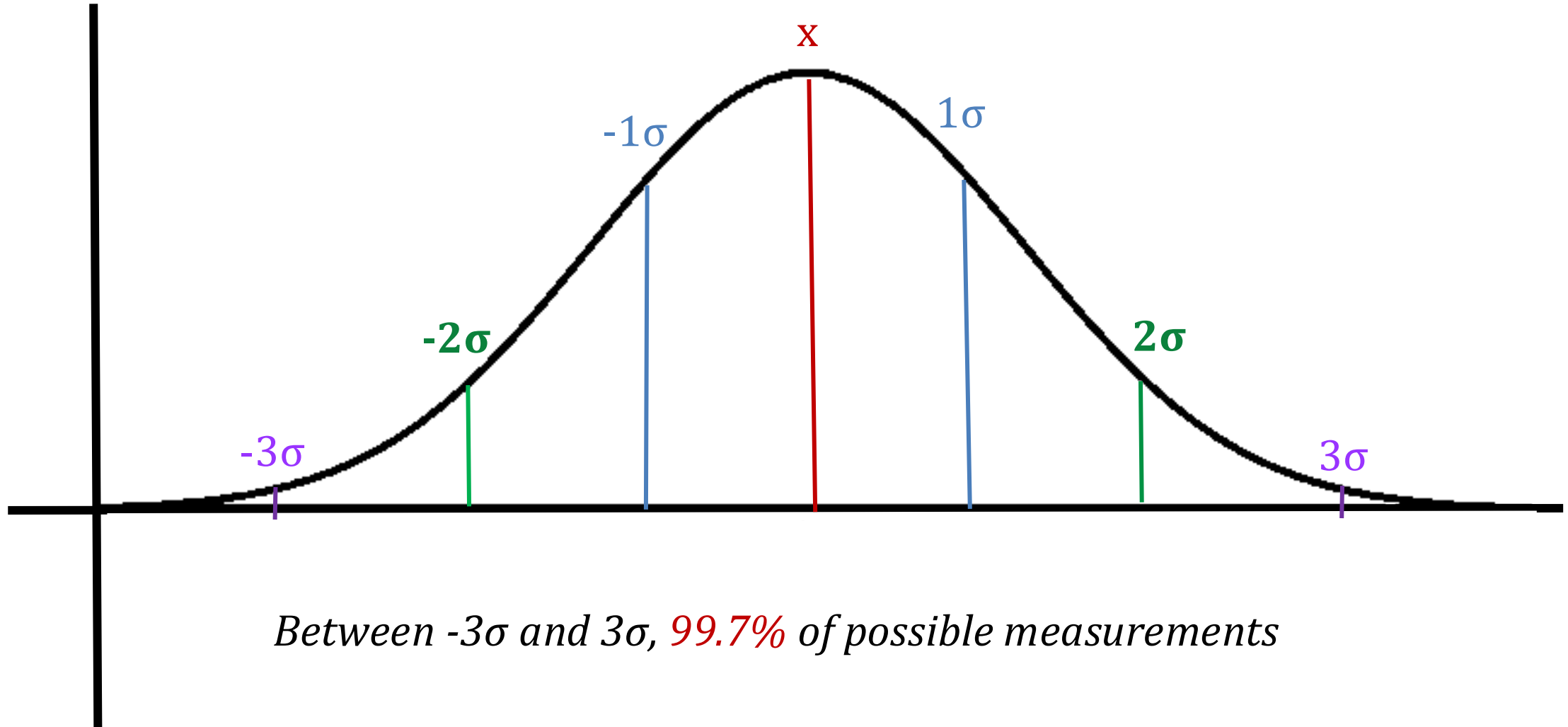


Between -1σ and 1σ , 68.3% of possible measurements

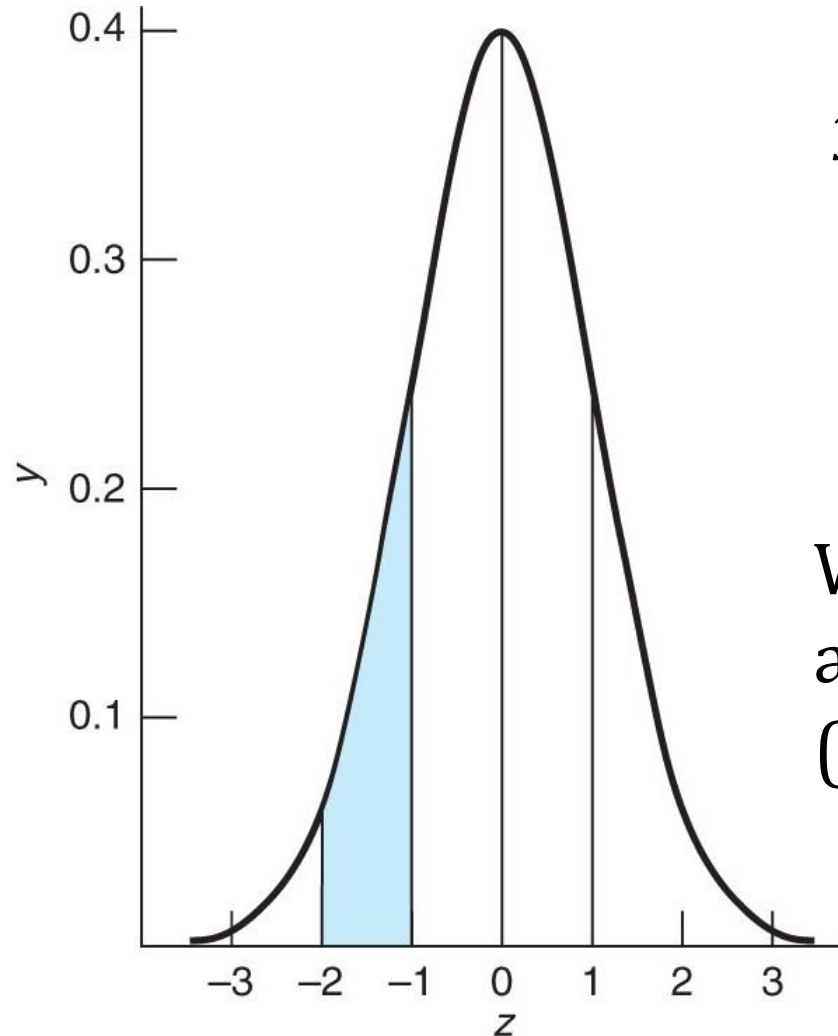
Standard Deviations



Standard Deviations



Gaussian Distributions



$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[-(x-\mu)^2/(2\sigma^2)\right]} \xrightarrow{\text{substitute } z = \frac{x-\mu}{\sigma}}$$

$$y = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

What is the likelihood data will fall between -1 and -2 standard deviations away from the mean?
(the area that is shaded in)

Gaussian Distributions

Ordinate and area for the normal (Gaussian) error curve, $y = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$

$ z ^a$	y	Area ^b	$ z $	y	Area	$ z $	y	Area
0.0	0.398 9	0.000 0	1.4	0.149 7	0.419 2	2.8	0.007 9	0.497 4
0.1	0.397 0	0.039 8	1.5	0.129 5	0.433 2	2.9	0.006 0	0.498 1
0.2	0.391 0	0.079 3	1.6	0.110 9	0.445 2	3.0	0.004 4	0.498 650
0.3	0.381 4	0.117 9	1.7	0.094 1	0.455 4	3.1	0.003 3	0.499 032
0.4	0.368 3	0.155 4	1.8	0.079 0	0.464 1	3.2	0.002 4	0.499 313
0.5	0.352 1	0.191 5	1.9	0.065 6	0.471 3	3.3	0.001 7	0.499 517
0.6	0.333 2	0.225 8	2.0	0.054 0	0.477 3	3.4	0.001 2	0.499 663
0.7	0.312 3	0.258 0	2.1	0.044 0	0.482 1	3.5	0.000 9	0.499 767
0.8	0.289 7	0.288 1	2.2	0.035 5	0.486 1	3.6	0.000 6	0.499 841
0.9	0.266 1	0.315 9	2.3	0.028 3	0.489 3	3.7	0.000 4	0.499 904
1.0	0.242 0	0.341 3	2.4	0.022 4	0.491 8	3.8	0.000 3	0.499 928
1.1	0.217 9	0.364 3	2.5	0.017 5	0.493 8	3.9	0.000 2	0.499 952
1.2	0.194 2	0.384 9	2.6	0.013 6	0.495 3	4.0	0.000 1	0.499 968
1.3	0.171 4	0.403 2	2.7	0.010 4	0.496 5	∞	0	0.5

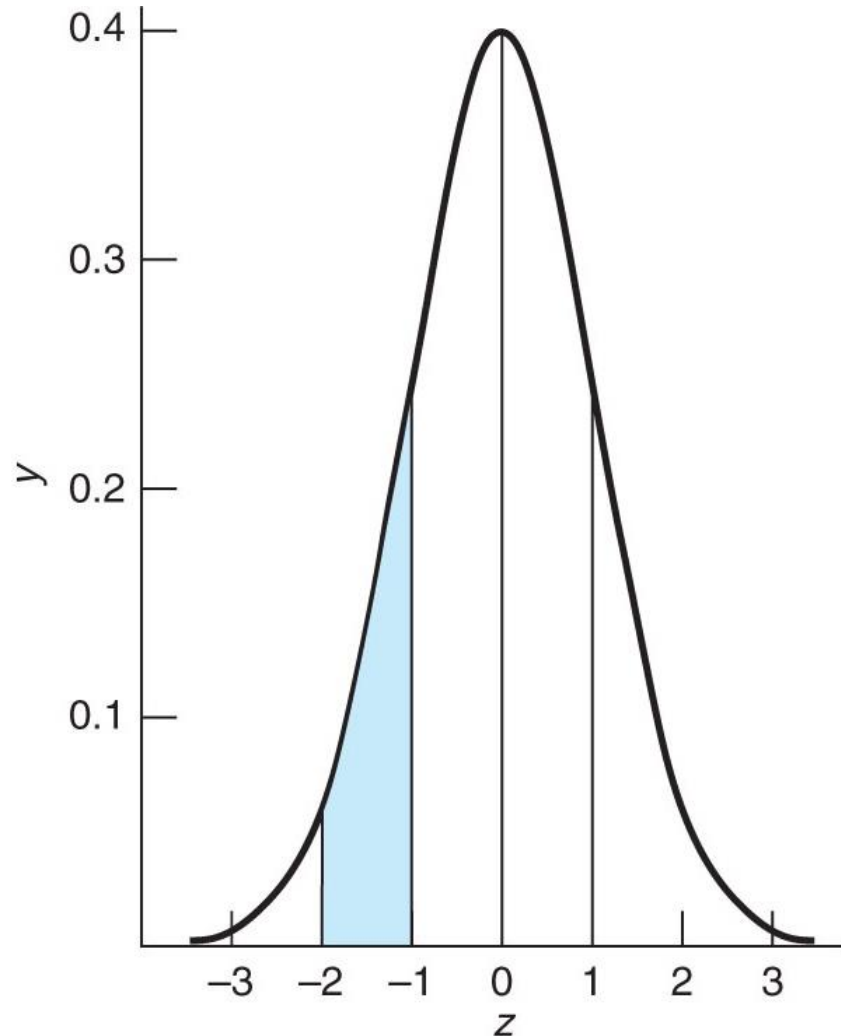
Key Notes

- z measures **how far a value is from the mean** in units of the standard deviation (σ). Larger z means farther from the mean.
- The table value y is the height of the curve (not the probability. Probability comes from the Area column i.e., area under the curve).
- From the table, **probability increases with z** (area from the mean outward):
 $z = 1 \rightarrow 0.3413$, $z = 2 \rightarrow 0.4773$, $z = 3 \rightarrow 0.4987$, $z = 4 \rightarrow \approx 0.5000$ (per side).
- At the same time, the curve height y drops rapidly:
 $y \approx 0.24$ ($z=1$) $\rightarrow 0.05$ ($z=2$) $\rightarrow 0.004$ ($z=3$) $\rightarrow \sim 0$ ($z=4$), showing extreme deviations are very rare.

^a $z = (x - \mu)/\sigma$.

^b The area refers to the area between $z = 0$ and $z =$ the value in the table. Thus the area from $z = 0$ to $z = 1.4$ is 0.419 2. The area from $z = -0.7$ to $z = 0$ is the same as from $z = 0$ to $z = 0.7$. The area from $z = -0.5$ to $z = +0.3$ is $(0.191 5 + 0.117 9) = 0.309 4$. The total area between $z = -\infty$ and $z = +\infty$ is unity.

Gaussian Distributions



From the table on the previous page

$$-2\sigma=0.4773$$

$$-1\sigma=0.3413$$

This represents the areas from 0σ to the number of standard deviations

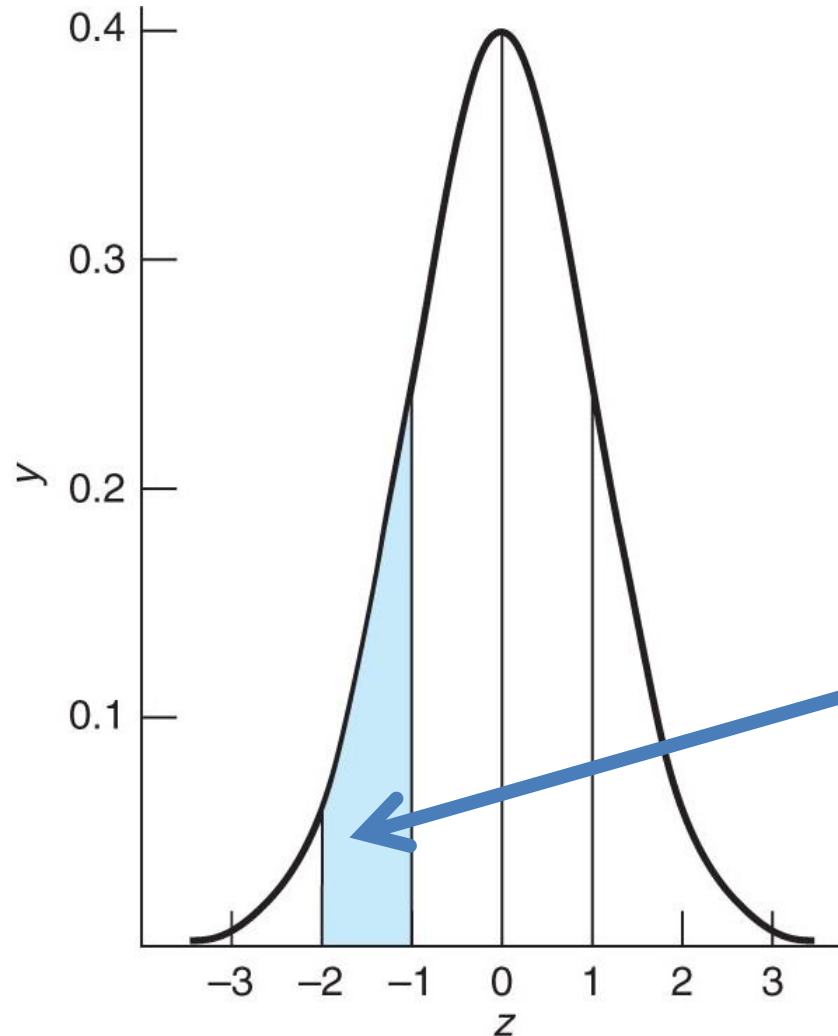
So, 0.4773 is the area from 0σ to -2σ (or 0σ to 2σ)

If we subtract the areas

$$0.4773-0.3413=0.136$$

Or 13.6%

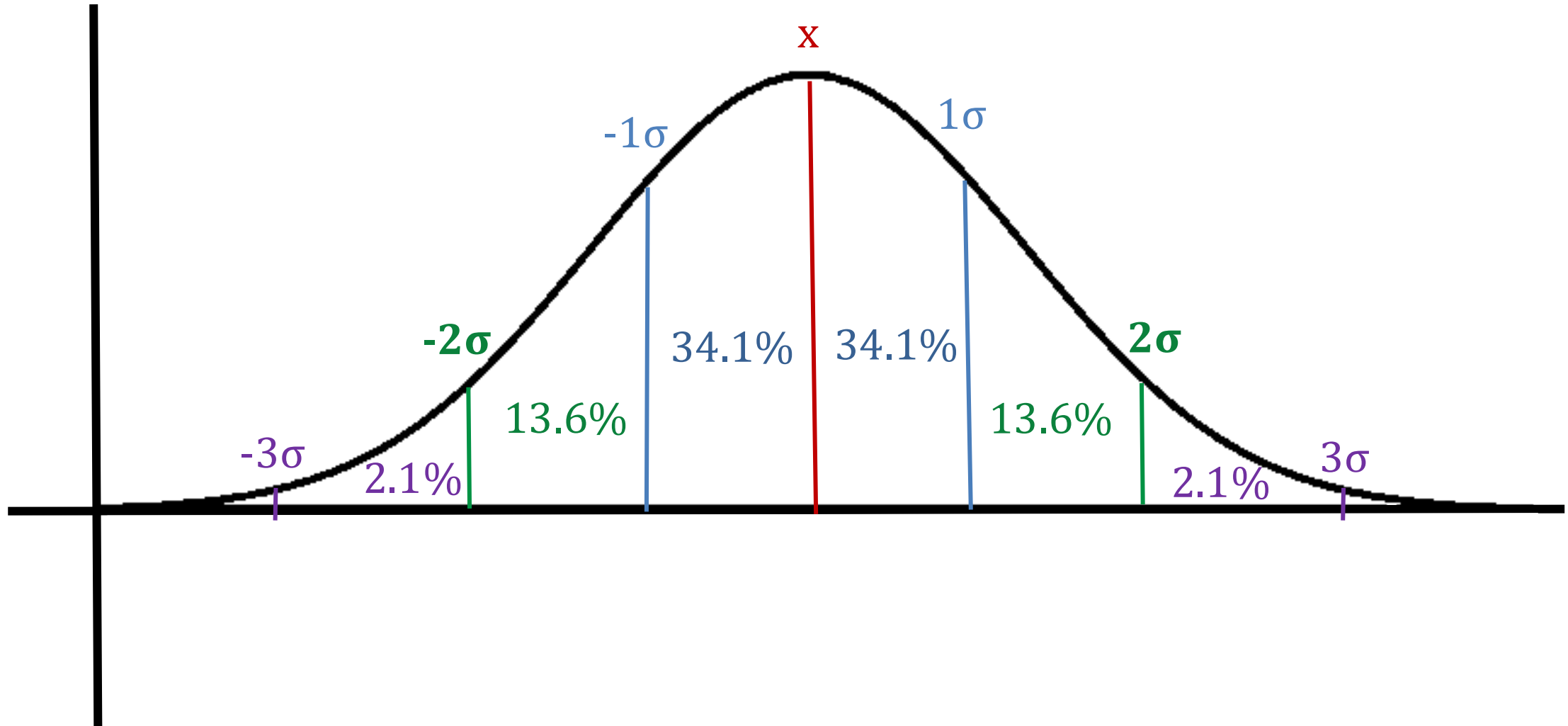
Gaussian Distributions



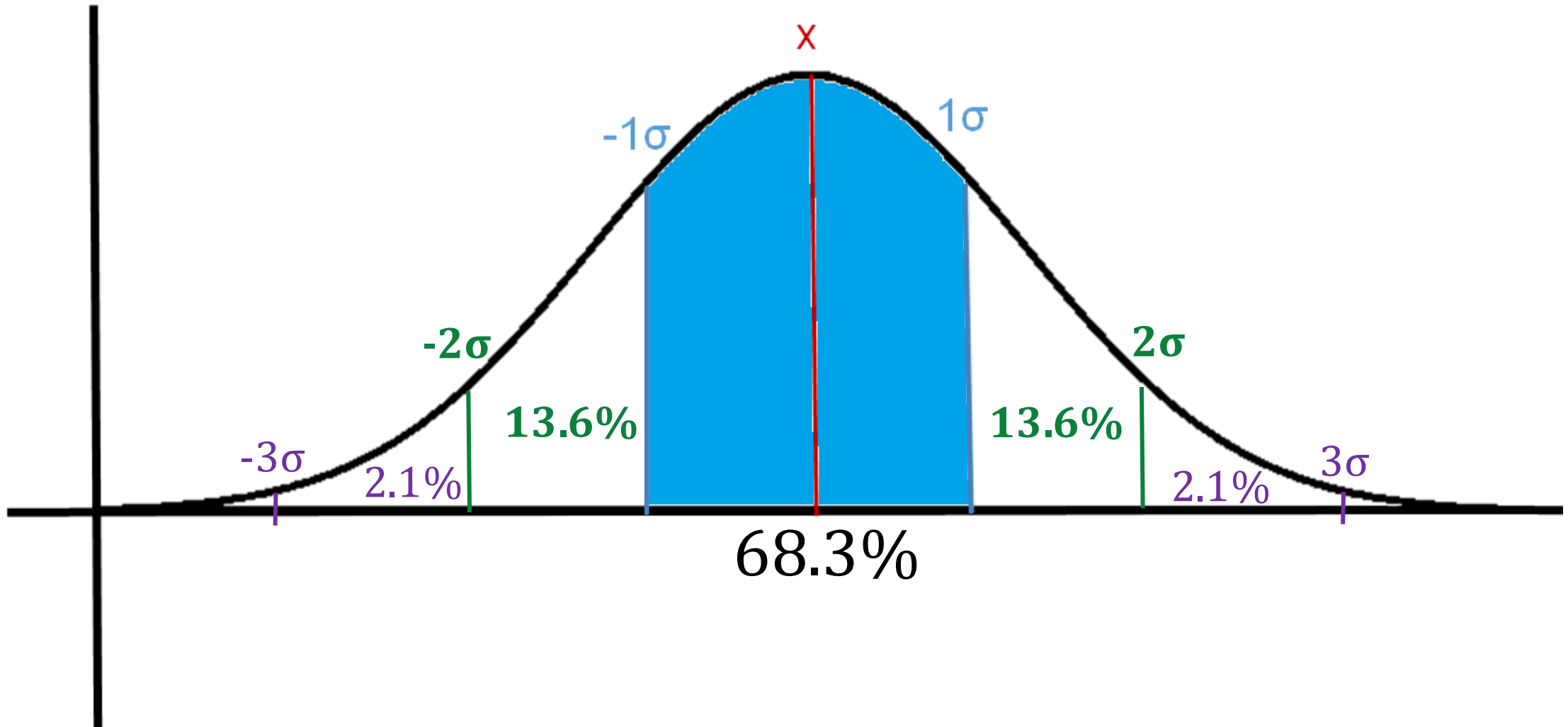
Area = 0.136

13.6% probability

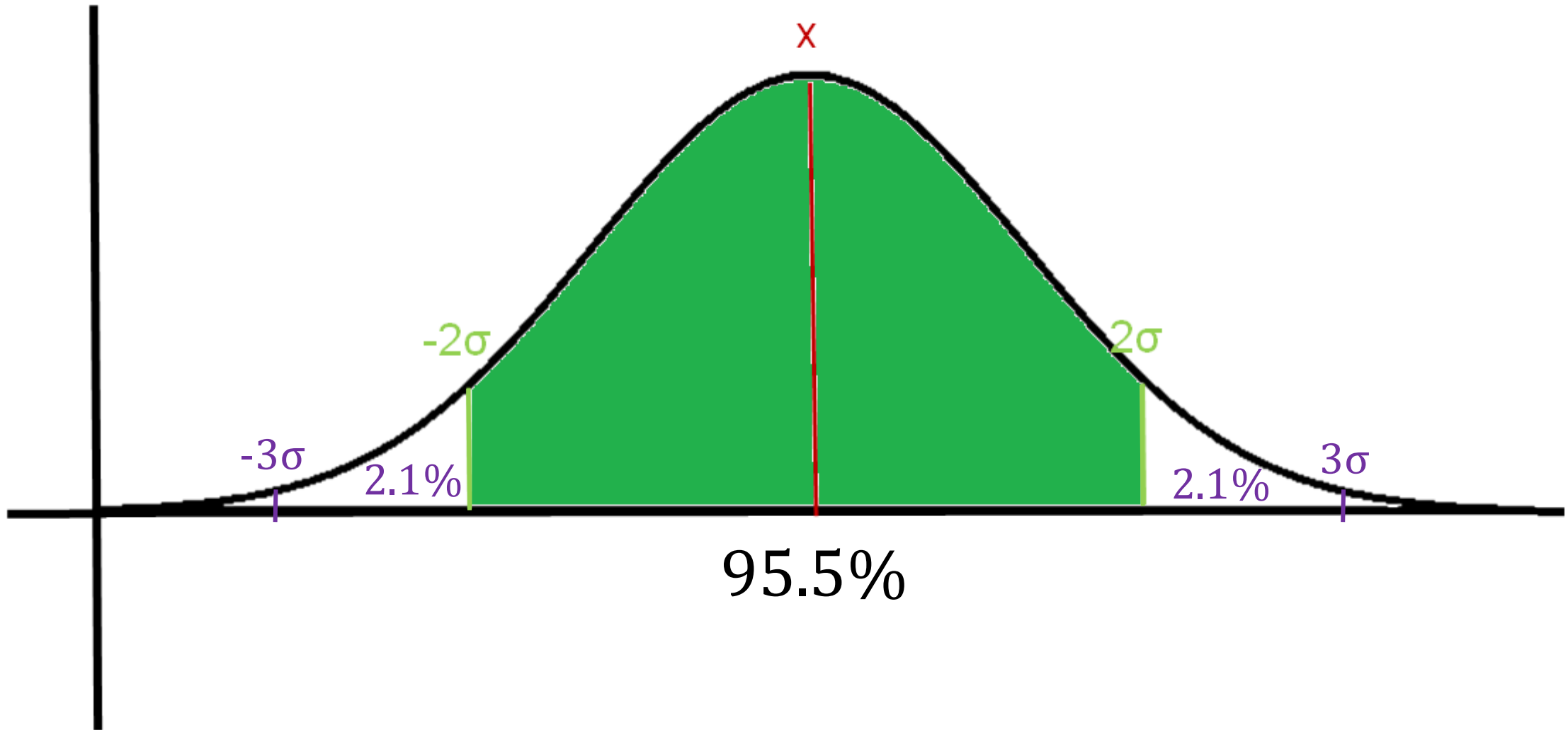
Gaussian Distributions



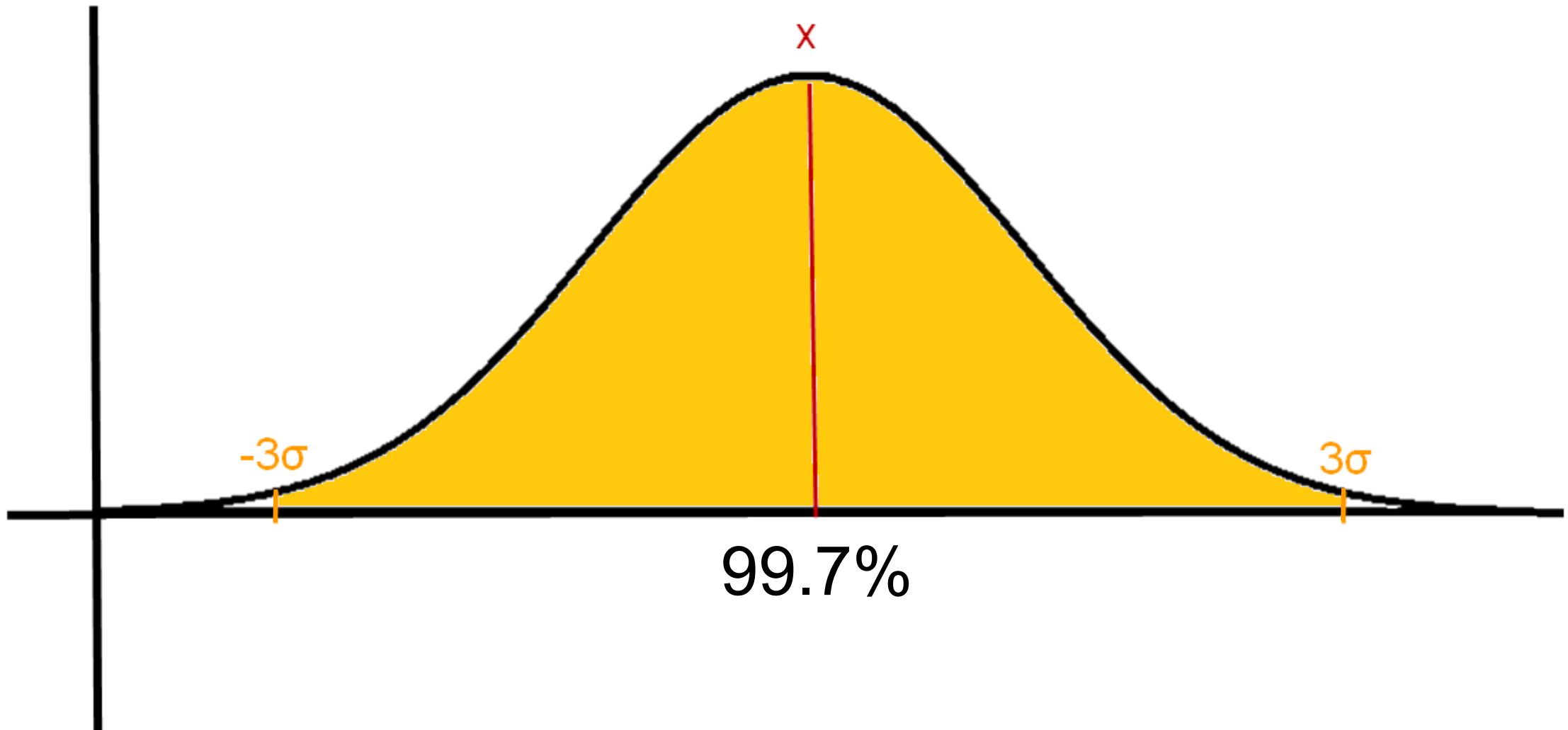
Gaussian Distributions



Gaussian Distributions

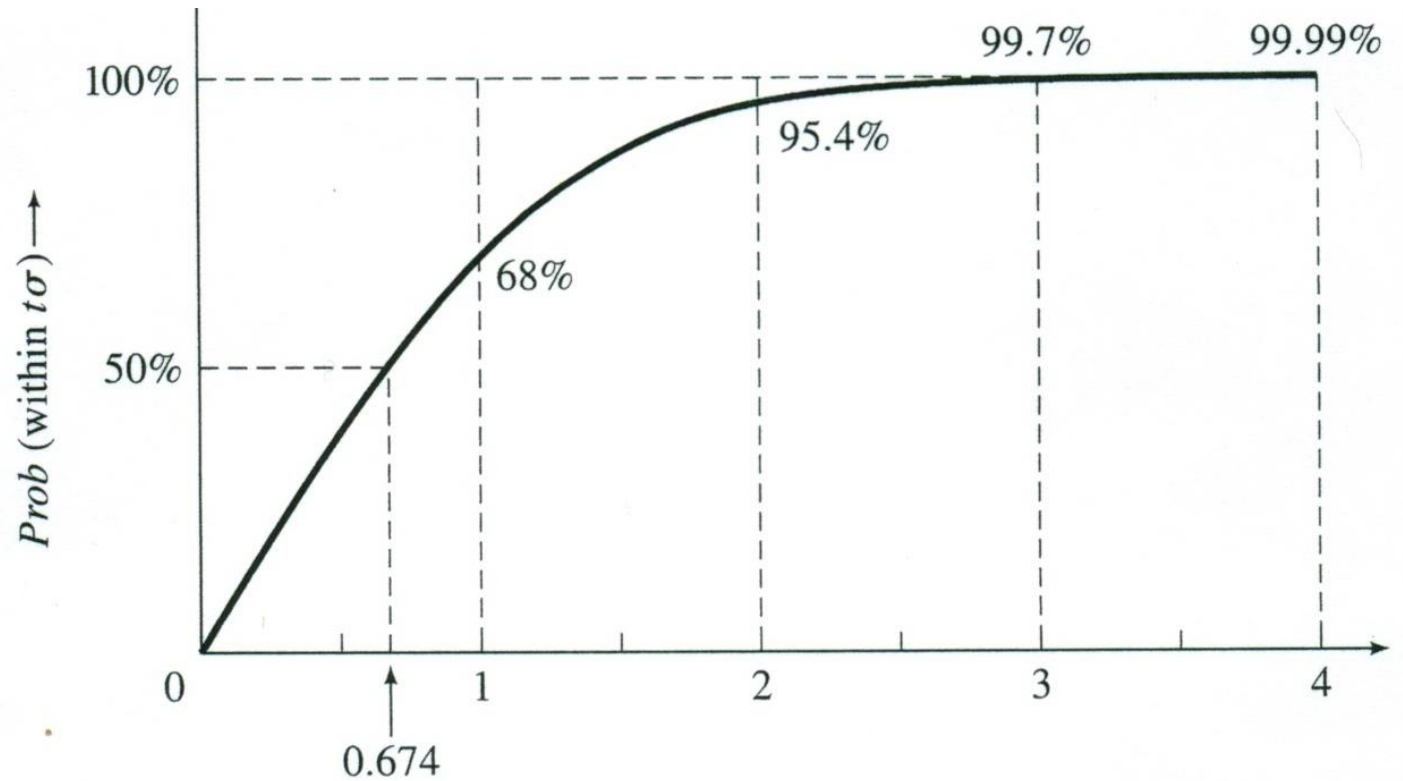


Gaussian Distributions



Probability vs Standard Deviation

Error Function, a.k.a. Normal Error Integral

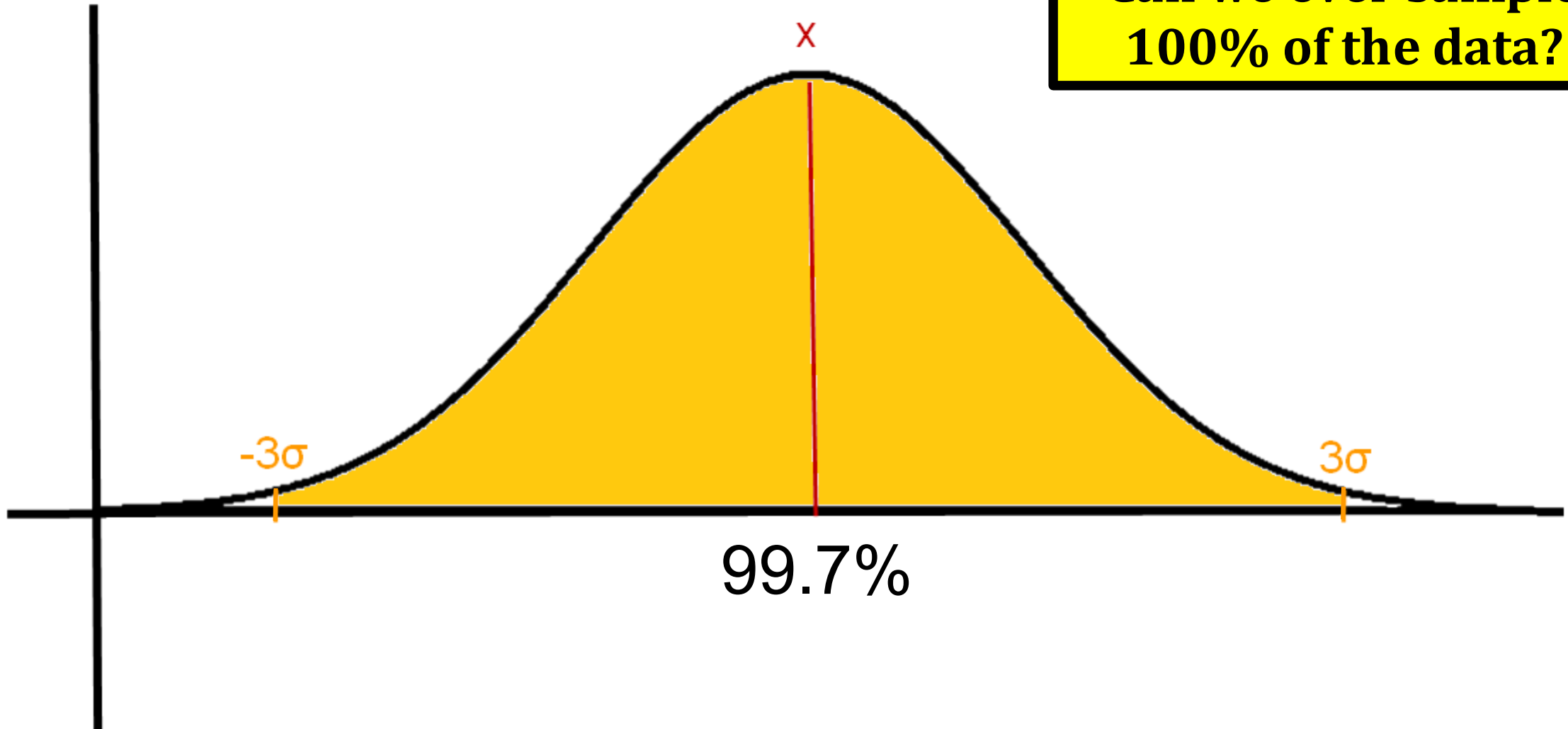


z
↓
 t

t	0	0.25	0.5	0.75	1.0	1.25	1.5	1.75	2.0	2.5	3.0	3.5	4.0
Prob (%)	0	20	38	55	68	79	87	92	95.4	98.8	99.7	99.95	99.99

Gaussian Distributions

Can we ever sample
100% of the data?



The **Sample** Standard Deviation

$$s = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{N - 1}}$$

This equation applies to small sets of data and represents the number of **degrees of freedom (N-1)**.

The number of degrees of freedom indicates the number of independent results that enter into the computation of the standard deviation.

Sample variance (s^2) is an estimate of the population variance (σ^2).

Degrees of Freedom

Days of Hat Freedom!



If you're averaging a data set of values, the same holds true.

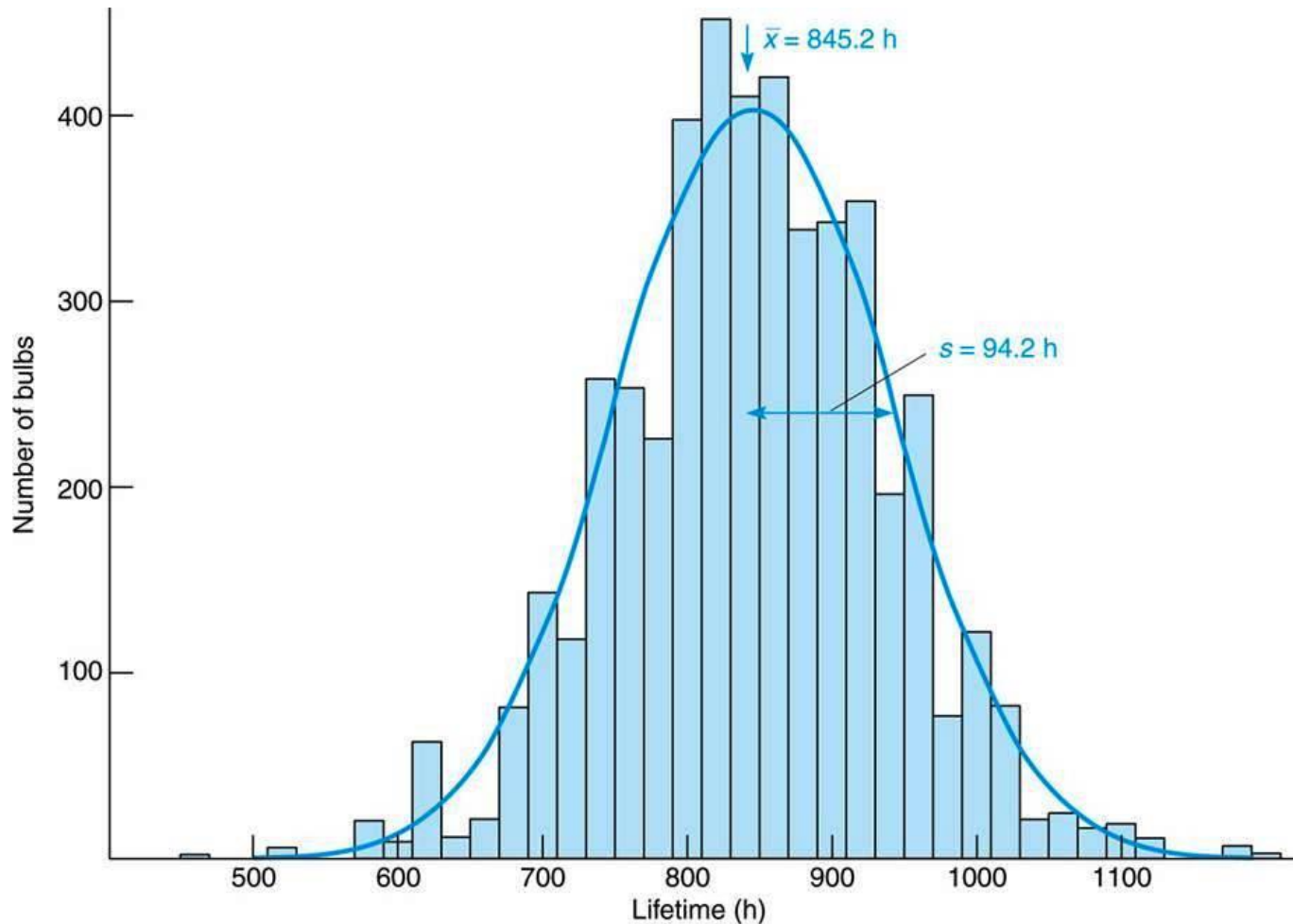
The sum of all values = number of values x mean

The first 6 values could be anything,
but the last must specifically make the sum calculation true
Hence, N-1

[Link](#)

Example: Lightbulb Lifetimes

What fraction of the bulbs will fail before 600 hours?

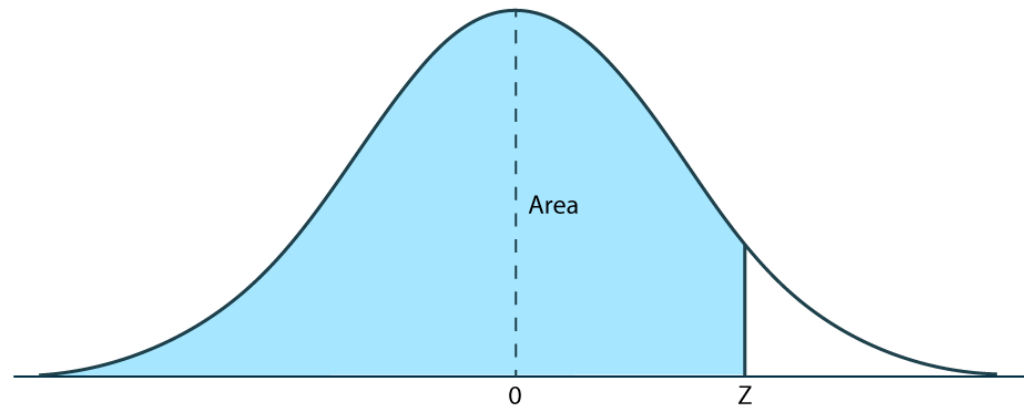


$$\bar{x} = 845.2 \text{ h}$$

$$s = 94.2 \text{ h}$$

$$z = \frac{x - \bar{x}}{s} = \frac{600 - 845.2}{94.2} = -2.60$$

Example: Lightbulb Lifetimes



Remember: the area from the mean to zero is 0.5000

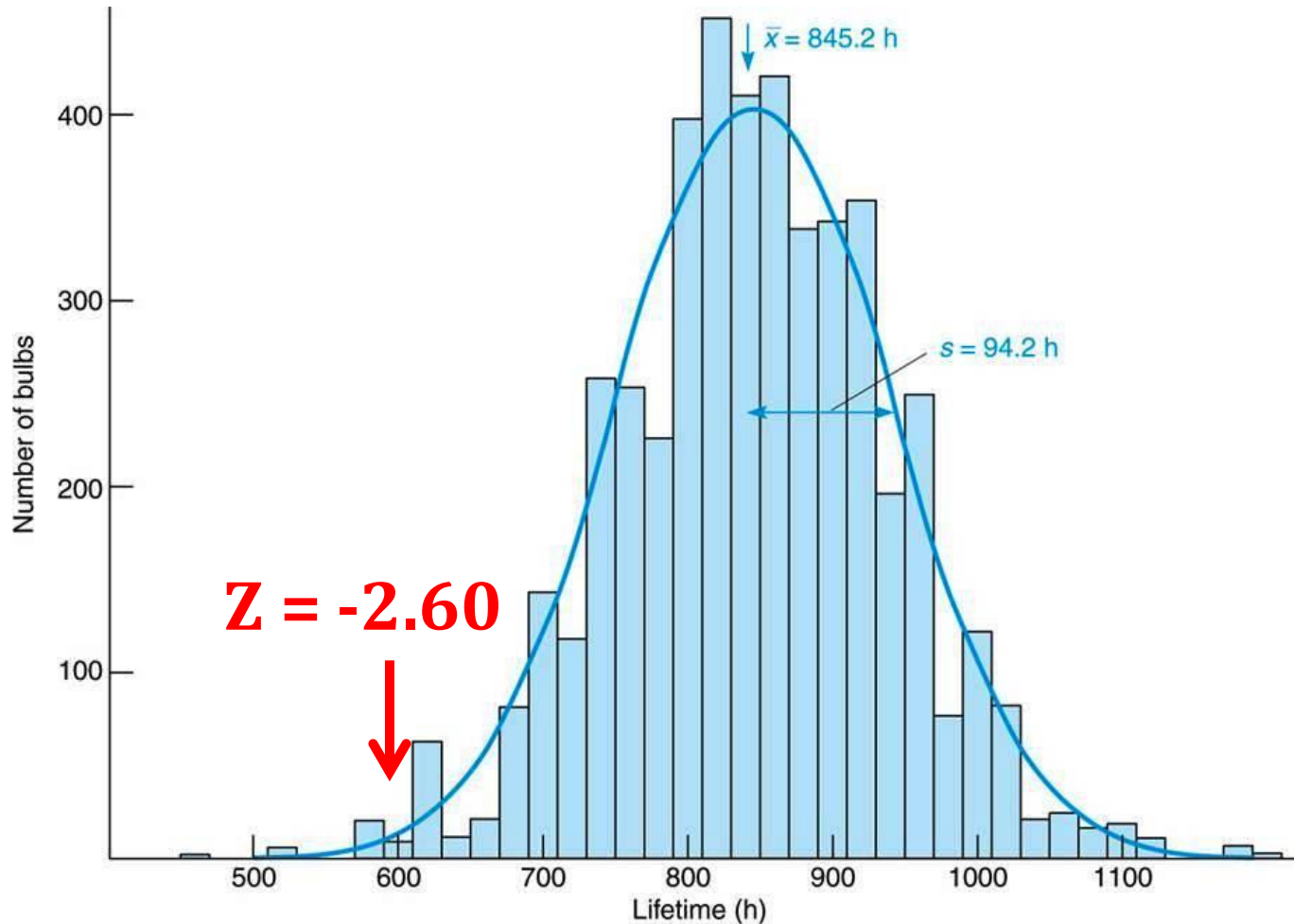
$$\begin{aligned} \text{Area} &= 0.4953 \quad (\text{from table}) \\ 0.5000 - 0.4953 &= 0.0047 \end{aligned}$$



z	y	Area
1.4	0.149 7	0.419 2
1.5	0.129 5	0.433 2
1.6	0.110 9	0.445 2
1.7	0.094 1	0.455 4
1.8	0.079 0	0.464 1
1.9	0.065 6	0.471 3
2.0	0.054 0	0.477 3
2.1	0.044 0	0.482 1
2.2	0.035 5	0.486 1
2.3	0.028 3	0.489 3
2.4	0.022 4	0.491 8
2.5	0.017 5	0.493 8
2.6	0.013 6	0.495 3
2.7	0.010 4	0.496 5

Example: Lightbulb Lifetimes

What fraction of the bulbs will fail before 600 hours?



$$\bar{x} = 845.2 \text{ h}$$

$$s = 94.2 \text{ h}$$

$$Z = \frac{x - \bar{x}}{s} = \frac{600 - 845.2}{94.2} = -2.60$$

0.47% of the bulbs are
expected to fail below 600
hours

Practice Problem

The following results were obtained in the replicate determination of the lead content of a blood sample: 0.752, 0.756, 0.752, 0.751, and 0.760 ppm Pb.

Find the mean and the standard deviation of this set of data.

Standard Error of the Mean

In a Gaussian distribution, there's a 95.4% chance that a single measurement will fall within ± 2 standard deviations of the mean.

If you take **several measurements and average them**, the resulting mean becomes more precise, with less scatter as the number of data points (N) increases.

This reduced spread is described by the **standard error of the mean (σ_m or s_m)**, which equals the standard deviation divided by the square root of N.

$$s_m = \frac{s}{\sqrt{N}}$$

Repeating this process across multiple sets can give a more reliable estimate of both the mean and the standard deviation.

Pooling Data to Improve the Reliability of s

- Pooling data collected using similar techniques is one way to improve the estimate of the population standard deviation.
- The weighted average of individual estimates of the standard deviation is called the **pooled standard deviation** or *s_{pooled}*

$$s_{pooled} = \sqrt{\frac{\sum_{i=1}^{N_1} (x_{i_1} - \bar{x}_1)^2 + \sum_{j=1}^{N_2} (x_{j_2} - \bar{x}_2)^2 + \sum_{k=1}^{N_3} (x_{k_3} - \bar{x}_3)^2 + \dots}{(N_1 - 1) + (N_2 - 1) + (N_3 - 1) + \dots}}$$

$$\text{Or } s_{pooled} = \sqrt{\frac{s_1^2 (N_1 - 1) + s_2^2 (N_2 - 1)}{N_1 + N_2 - 2}}$$

See EXAMPLE 4-2 of textbook

Other Measures of Precision

Variance (s^2): is the square of the standard deviation

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{N - 1}$$

- Note that the standard deviation has the **same units as the data**, but the variance has the **units of the data squared** (e.g., ppm²)."
- It is an estimate of the population variance (σ^2)

Other Measures of Precision

Relative Standard Deviation (RSD) and Coefficient of Variation (CV)

Frequently standard deviations are expressed in relative rather than absolute terms.

$$RSD = \frac{s}{\bar{x}}$$

CV: The RSD multiplied by 100% (expressed as percentage)

$$CV = \frac{s}{\bar{x}} \times 100\%$$

RSDs often give a clearer picture of data quality than do absolute standard deviations.

For a copper sample with a 2 mg standard deviation:

$$\text{At 50 mg mean} \rightarrow CV = \frac{2}{50} \times 100\% = 4\%$$

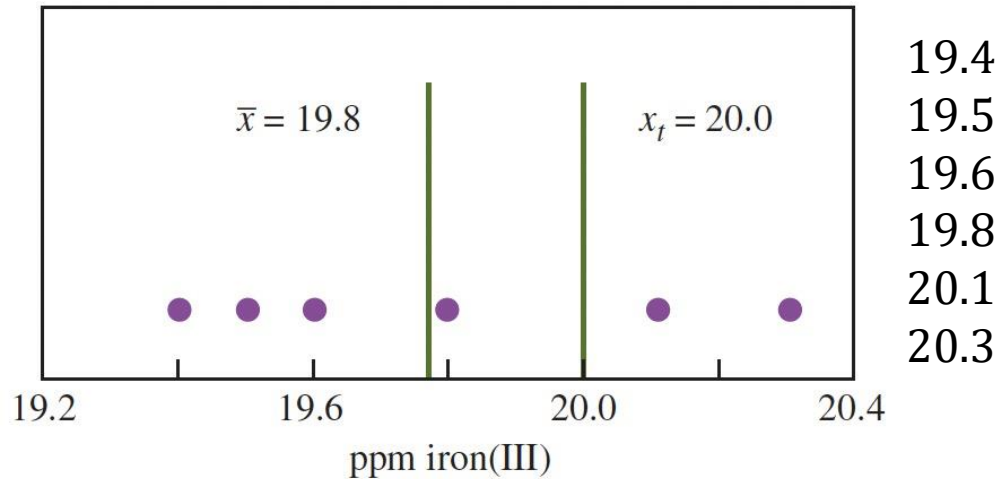
$$\text{At 10 mg mean} \rightarrow CV = \frac{2}{10} \times 100\% = 20\%$$

Smaller means lead to higher relative uncertainty.

Other Measures of Precision

Range (w)

The difference between the **largest value in the set and the smallest**: is used to describe the precision of a set of replicate results



$$w = 20.3 - 19.4 = 0.9 \text{ ppm Fe}$$

Standard Deviation of Calculated Results

Often we must estimate the standard deviation of a result that has been calculated from two or more experimental data points, each of which has a known sample standard deviation.

When results are combined, error propagation occurs.

TABLE 4-4

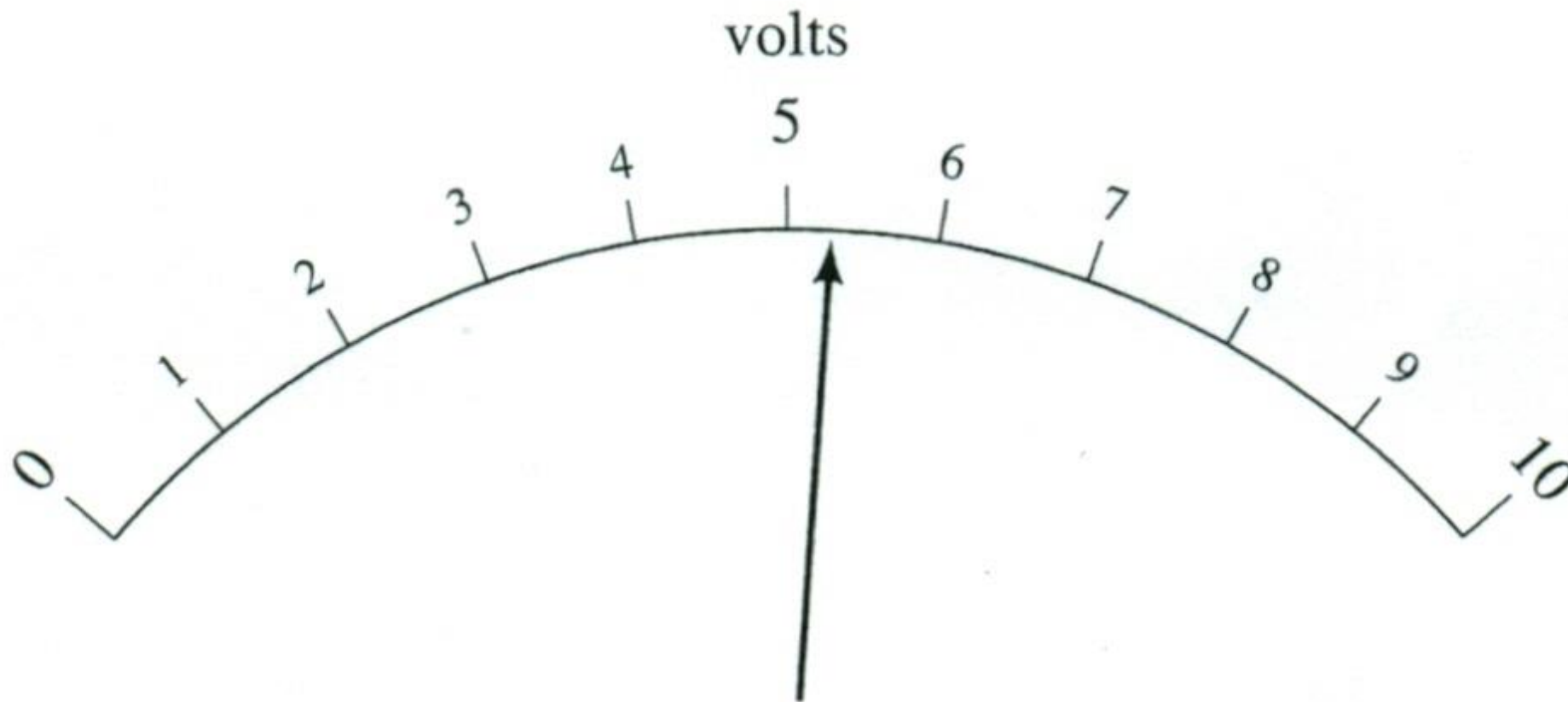
Error Propagation in Arithmetic Calculations		
Type of Calculation	Example*	Standard Deviation of $y^\dagger$
Addition or subtraction	$y = a + b - c$	$s_y = \sqrt{s_a^2 + s_b^2 + s_c^2}$ (1)
Multiplication or division	$y = a \times b/c$	$\frac{s_y}{y} = \sqrt{\left(\frac{s_a}{a}\right)^2 + \left(\frac{s_b}{b}\right)^2 + \left(\frac{s_c}{c}\right)^2}$ (2)
Exponentiation	$y = a^x$	$\frac{s_y}{y} = x\left(\frac{s_a}{a}\right)$ (3)
Logarithm	$y = \log_{10} a$	$s_y = 0.434 \frac{s_a}{a}$ (4)
Antilogarithm	$y = \text{antilog}_{10} a$	$\frac{s_y}{y} = 2.303 s_a$ (5)

* a , b , and c are experimental variables with standard deviations of s_a , s_b , and s_c , respectively.

[†]These relationships are derived in Appendix 9. The values for s_y/y are absolute values if y is a negative number.

We will get back to this shortly!

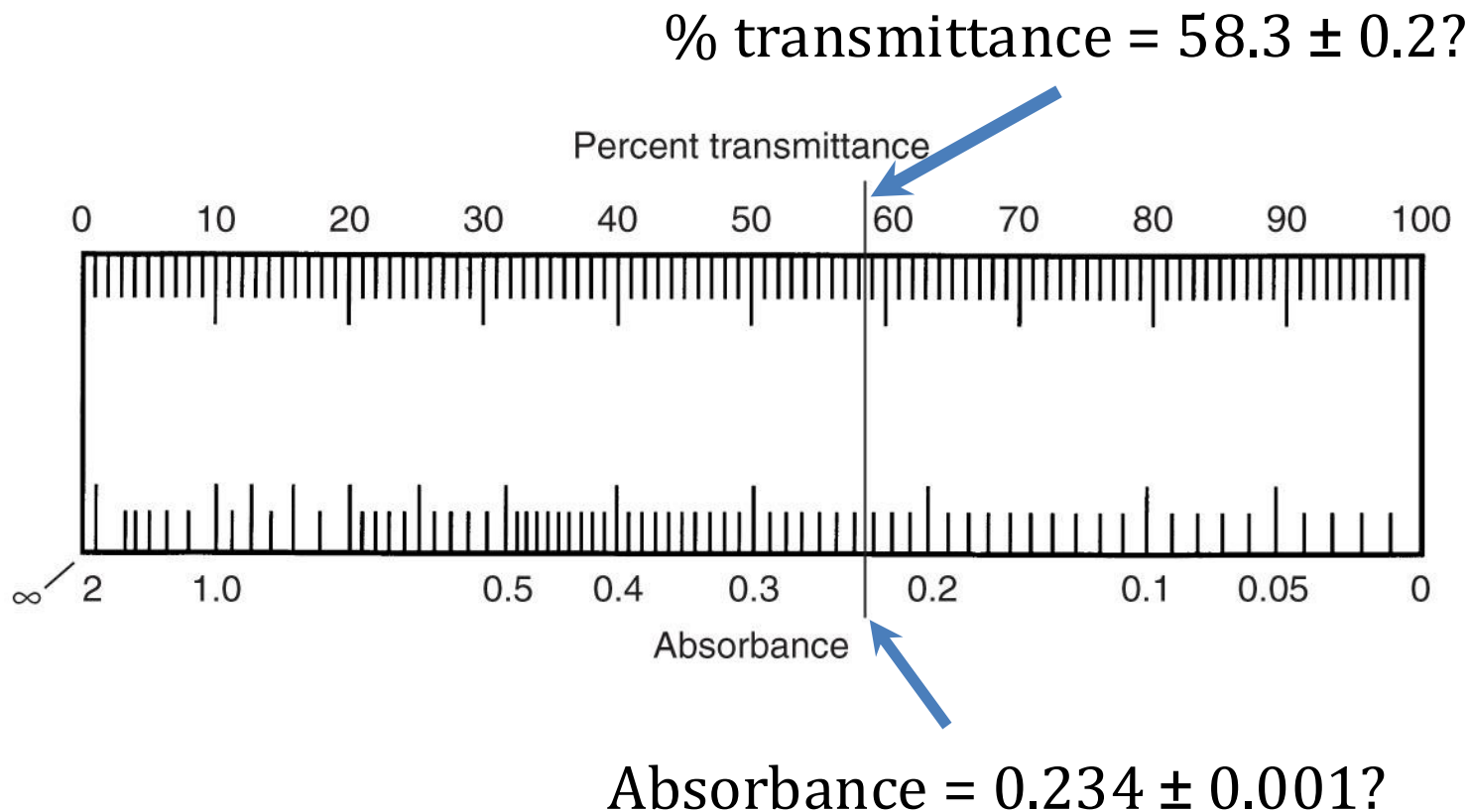
Recall: Reading a Scale



Best estimate of the voltage = 5.3 volts
Probable range = **5.2 – 5.4** volts

Reading a Scale

The approximate estimation of uncertainty is fairly easy when the only problem is to locate a point on a marked scale. Imagine if the needle were moving, noisy, oscillating, etc.



Significant Figures

Throughout this course, you are going to be seeing a lot of numbers
(especially in lab!)

As simple as it would be to just perform the operations to get the final answer, that final answer **isn't really final...**

The uncertainty sets the significant figures!

Multiple measurements can **will not be exactly the same** due to errors.

We then want to report the data in the most precise manner possible.

Significant Figures (Sig Figs): the number of digits needed to give the correct information without losing any precision

Significant Figures

Any **nonzero** digit is a significant figure (sig figs)

23923

has 5 sig figs

7.421

has 4 sig figs

Scientific notation does not change the number of sig figs.

42324 and 4.2324×10^4

both have 5 sig figs

Ambiguity: 92500 → trailing zeros without a decimal point may only indicate the location of the decimal point.

9.25×10^4

has 3 sig figs

9.250×10^4

has 4 sig figs

9.2500×10^4

has 5 sig figs

Significant Figures

Zeros: Sometimes they are significant, sometimes they aren't

Zeros in the middle of a number **are significant**.

303, 2301, 2.304×10^5

Zeros at the end of a number **are significant** if a decimal point is explicitly shown.

330., 24420., 2.330×10^5

Trailing zeros **are significant** if they appear after a decimal point..

Not the same answer:

3.3×10^2 3.30×10^2

Significant Figures

Zeros: Sometimes they are significant, sometimes they aren't

Zeros at the start of a number **are not significant**, even if a decimal point is present. Leading zeros are **placeholders** used only to locate the decimal point.

0.4**0**2

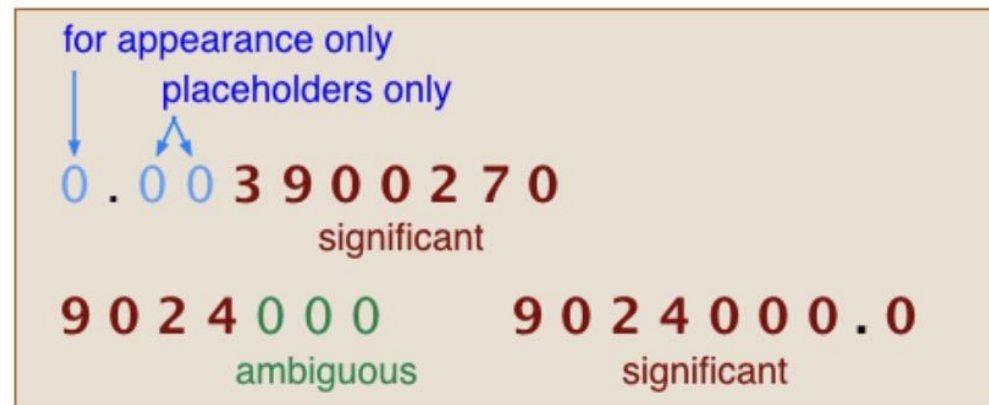
has 3 sig figs

0.000000000000001

has 1 sig fig

4**0**2 and $0.4**0**2 \times 10^3$

both have 3 sig figs



Significant Figures

What determines the number of sig figs in a measurement?

The way the measurement was made.

- Each instrument, device, glassware etc. all have different levels of significance
- Exact numbers (counting numbers and defined constants) have infinite significant figures and do not limit the precision of a calculation.

Rounding Data

Rounding Data

When rounding a number to **n significant digits**:

- The **most significant digit** is the **left-most nonzero digit**.
- The **least significant digit** is the **nth digit from the most significant digit**.
- The **first non-significant digit** is the **(n + 1)th digit**.

Example: rounding **0.003742** to 3 significant digits

- Most significant digit → **3**
- Least significant digit → **7**
- First non-significant digit → **4**

(Used to decide whether rounding occurs)

Rounding Data

Rounding Data

Let the $(n + 1)$ th digit be the first non-significant digit.

- If it is **less than 5** → the last significant digit **remains unchanged**.
- If it is **greater than 5** → the last significant digit is **increased by 1**.
- If it is **exactly 5**:
 - **No nonzero** digits follow → round to the **nearest even digit**.
 - **Any nonzero** digit follows → **round up**.
- Remove **all non-significant** digits.
- Do not round intermediate calculations — **round only the final result**.

Rounding Data

Example: round these off to 3 significant figures.

Example	Answer	Reason
34.216	34.2	First non-significant digit (1) is less than 5, digits after the third significant figure are dropped.
39.99	40.0	The 4th digit (9) is ≥ 5 , so the 3rd digit rounds up, giving 40.0 (3 significant figures).
85,381	<u>85,400</u>	The trailing zeros indicate the rounded place value (placeholders)
0.04597	0.0460	Leading zeros are placeholders and are not significant.

Example: round the final result to 3 significant figures
 $(1.234 \times 2.346)=?$

Correct procedure (do NOT round intermediate values):

$$1.234 \times 2.346 = 2.894964 \rightarrow 2.89$$

Incorrect procedure (rounding too early):

$$1.23 \times 2.35 = 2.8905 \rightarrow 2.90$$

Rounding Data

Example: round these off to 3 significant figures:

Number	Rounded	Reason
2.334765	2.33	next digit <5
2.338766	2.34	next digit >5
2.335001	2.34	5 followed by nonzero
2.335000	2.34	5 exactly → even
2.345000	2.34	4 is even
2.345002	2.35	5 followed by nonzero

*Consider all digits when rounding.
If the first discarded digit is exactly 5
and no nonzero digits follow, round
to the nearest even digit.*

*This prevents systematic rounding
bias.*

Rounding Data with Uncertainty

How to round: 3.98 ± 0.05 g?

$$3.98 - 0.05 = 3.93 \text{ g}$$

$$3.98 + 0.05 = 4.03 \text{ g}$$

The value 3.98 suggests precision to the hundredths place, but the **uncertainty (± 0.05 g)** means the hundredths digit is uncertain.

Option 1: Strict interpretation

- Over-rounding (not recommended): rounding to 1 sig fig exaggerates uncertainty.
- Final result: 4 g: this implies a wide range: ± 0.5 g, which is a huge uncertainty compared to the original ± 0.05 g.

Option 2: More nuanced interpretation

- Keep 2 significant figures $\rightarrow 4.0$ g
- This implies an uncertainty of ± 0.05 g, which is much closer to the actual precision you had.

Rounding Data with Uncertainty

How to round: 3.98 ± 0.05 g?

Rounded Value	Implied Max	Implied Min	Absolute Uncertainty	Relative Uncertainty
3.98 g	3.985 g	3.975 g	± 0.005 g (rounded to 0.01 g)	1 in 400, or 0.25%
4 g	4.5 g	3.5 g	± 0.5 g or 1 g	1 in 4, 25%
4.0 g	4.05 g	3.95 g	± 0.05 g or 0.1 g	1 in 40, 2.5%

- Round the value so its last digit aligns with the uncertainty; that is, the number of digits (counting from the left) that are **known exactly, plus one** more.
- When this cannot be applied, then we round the result, so the relative uncertainty (uncertainty $\div$ value) stays close to the original.

Rounding Data with Uncertainty

The uncertainty sets the significant figures

$$6.3456788 \pm 0.00345762 \text{ g}$$

Must be written as:

$$6.346 \pm 0.003 \text{ g}$$

The first uncertain digit is the last significant figure.

Significant Figures: Math Time

How sig figs carry through to a final answer is completely dependent on which operation you are performing.

Addition and Subtraction

Multiplication and Division

Logarithms

Significant Figures: Addition and Subtraction

Align your numbers in their appropriate places (100s place, 10s place, 1s place, etc.) and perform the arithmetic operation.

If the numbers have the **same number of decimal places**, then the **sigfigs simply carry through**

$$\begin{array}{r} 1.362 \\ +3.111 \\ \hline 4.473 \end{array}$$

$$\begin{array}{r} 5.345 \\ +6.728 \\ \hline 12.073 \end{array}$$

$$\begin{array}{r} 7.26 \\ -6.69 \\ \hline 0.57 \end{array}$$

Significant Figures: Addition and Subtraction

If the number of decimal places is not the same, still align the numbers in the appropriate places and perform the operation. The result is rounded to match the number with the **fewest decimal places**. Rounding rules apply.

Digits after the decimal sets the limit

$$\begin{array}{r} 12.345 \\ +1.2 \\ \hline 13.545 \rightarrow 13.5 \end{array}$$

1.2 has only 1 decimal place →
result must match that.

$$\begin{array}{r} 123.45 \\ +6.789 \\ \hline 130.239 \rightarrow 130.24 \end{array}$$

123.45 has 2 decimal places → keep 2.

Significant Figures: Multiplication and Division

The simplest of them all.

Perform the operation, and the result must contain the same number of significant figures as the measurement with the **fewest significant figures**.

$$\begin{array}{r} 3.26 \\ \times 1.78 \\ \hline 5.8028 \\ =5.80 \end{array}$$

$$\begin{array}{r} 4.3179 \\ \times 3.6 \\ \hline 15.5444 \\ =16. \end{array}$$

$$\begin{array}{r} 34.60 \\ \times 2.46287 \\ \hline 140486 \\ =14.05 \end{array}$$

The number with the fewest significant figures limits the precision of the result

Significant Figures: Logarithms and Antilogarithms

Answers in logs are divided into two parts: Characteristic and Mantissa

$$\text{Log } 339 = 2.530$$

Characteristic

Mantissa

The number of digits in the mantissa equals the number of significant figures **in the original value.**

$$10^{2.531} = 340 \quad (339.6)$$

$$10^{2.530} = 339 \quad (338.8)$$

$$10^{2.529} = 338 \quad (338.1)$$

Significant Figures: Logarithms and Antilogarithms

$$\text{antilog}(-3.42) = 10^{-3.42} = 3.8 \times 10^{-4}$$

$$\log 0.001237 = -2.9076$$

$$\text{antilog} 4.37 = 2.3 \times 10^4$$

$$\log 1237. = 3.0924$$

$$10^{4.37} = 2.3 \times 10^4$$

Significant figures are determined by the mantissa

Practice Problem: Significant Figures

A) $1.362 \times 10^{-4} + 3.111 \times 10^{-4}$ $= 4.473 \times 10^{-4}$

B) $7.26 \times 10^{14} - 6.69 \times 10^{14}$ $= 0.57 \times 10^{14}$ or 5.7×10^{13}

C) $1.632 \times 10^5 + 4.107 \times 10^3 + 0.984 \times 10^6$ $= 11.51 \times 10^5$

D) $3.26 \times 10^{-5} \times 1.78$ $= 5.80 \times 10^{-5}$

E) $34.60 / 2.46287$ $= 14.05$

F) $\log(0.001237)$ $= 1.237 \times 10^{-3} \rightarrow \log(1.237) - 3 = -2.9076$

G) $10^{-2.600}$ $= 2.51 \times 10^{-3}$

Recall: Standard Deviation of Calculated Results

Often we must estimate the standard deviation of a result that has been calculated from two or more experimental data points, each of which has a known sample standard deviation.

When results are combined, error propagation occurs.

TABLE 4-4

Error Propagation in Arithmetic Calculations		
Type of Calculation	Example*	Standard Deviation of $y^\dagger$
Addition or subtraction	$y = a + b - c$	$s_y = \sqrt{s_a^2 + s_b^2 + s_c^2}$ (1)
Multiplication or division	$y = a \times b/c$	$\frac{s_y}{y} = \sqrt{\left(\frac{s_a}{a}\right)^2 + \left(\frac{s_b}{b}\right)^2 + \left(\frac{s_c}{c}\right)^2}$ (2)
Exponentiation	$y = a^x$	$\frac{s_y}{y} = x\left(\frac{s_a}{a}\right)$ (3)
Logarithm	$y = \log_{10} a$	$s_y = 0.434 \frac{s_a}{a}$ (4)
Antilogarithm	$y = \text{antilog}_{10} a$	$\frac{s_y}{y} = 2.303 s_a$ (5)

* a , b , and c are experimental variables with standard deviations of s_a , s_b , and s_c , respectively.

† These relationships are derived in Appendix 9. The values for s_y/y are absolute values if y is a negative number.

Relative vs. Absolute Uncertainty

$$12.35 \pm 0.02 \text{ mL}$$

$$\text{Relative uncertainty} = \frac{\text{absolute uncertainty}}{\text{measured value}} = \frac{0.02 \text{ mL}}{12.35 \text{ mL}} = 0.00162$$

$$\text{and Percent uncertainty} = 100 \times 0.00162 = 0.162\%$$

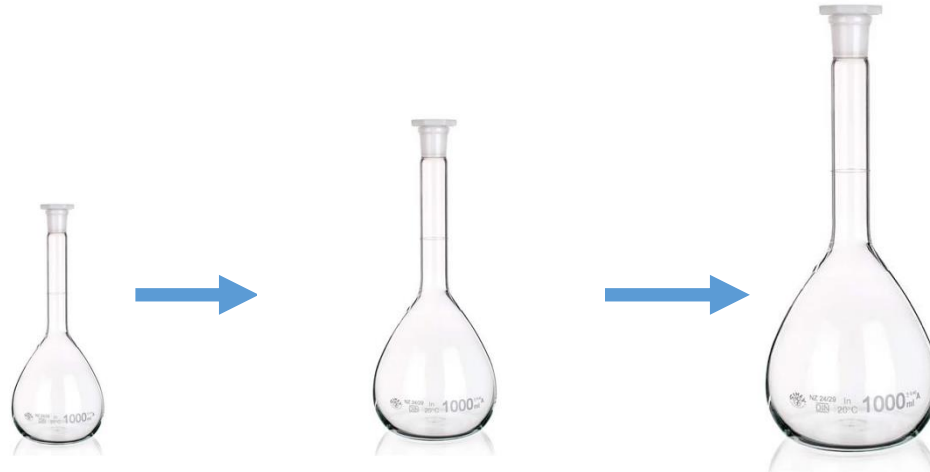
$$200 \pm 2 \qquad 1\%$$

$$5000 \pm 50 \qquad 1\%$$

Propagation of Error

Regardless of where the error comes from, it either needs to be corrected or accounted for in your final answer

Systematic error should be corrected when possible; random error must be propagated.

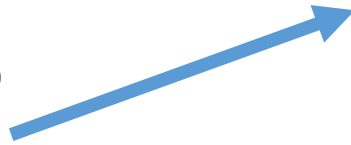


Random error needs to be accounted for in the final answer. To account for random error, we have to **propagate the error (or uncertainty)** from all the math that was performed

Standard Deviation of a Sum or Difference

Consider the sum:

$$\begin{array}{rcl} +0.5 & (\pm 0.02) & \\ +4.1 & (\pm 0.03) & \\ \hline -1.97 & (\pm 0.05) & \\ 2.63 & (?) & \end{array}$$



$$\begin{aligned} +0.02 + 0.03 + 0.05 &= +0.1 \\ -0.02 - 0.03 - 0.05 &= -0.1 \\ -0.02 - 0.03 + 0.05 &= 0 \\ +0.02 + 0.03 - 0.05 &= 0 \end{aligned}$$



The standard deviation of the sum will lie between these two extremes.

The variance of a sum or difference equals the sum of the variances of the individual terms.

$$s_y^2 = s_a^2 + s_b^2 + s_c^2 \quad \rightarrow \quad s_y = \sqrt{s_a^2 + s_b^2 + s_c^2}$$

$$s_y = \sqrt{(0.02)^2 + (0.03)^2 + (0.05)^2} = \pm 0.06$$

$$2.63 \pm 0.06$$

Standard Deviation of a Multiplication and Division

Consider the following computation:

$$\frac{4.10 (\pm 0.02) \times 0.005 (\pm 0.0001)}{1.97 (\pm 0.04)} = 0.010406 (\pm ?)$$

For multiplication or division, the relative standard deviation (RSD) of the result is the **square root of the sum of the squares of the relative standard deviations**.

$$\frac{s_y}{y} = \sqrt{\left(\frac{s_a}{a}\right)^2 + \left(\frac{s_b}{b}\right)^2 + \left(\frac{s_c}{c}\right)^2} \quad \frac{s_y}{y} = \sqrt{\left(\frac{0.02}{4.10}\right)^2 + \left(\frac{0.0001}{0.005}\right)^2 + \left(\frac{0.04}{1.97}\right)^2} = \pm 0.0289$$

To find the absolute SD: $s_y = y (= 0.010406) \times (\pm 0.0289) = \pm 0.000301$

$$0.0104 \pm 0.0003$$

See example 4-4

Standard Deviations in Exponential Calculations

Consider the relationship

$$y = a^x$$

y: result

a: base

x: exponent (can be considered to be free of uncertainty)

s_y : uncertainty in y

s_a : uncertainty in a

The relative standard deviation in y resulting from the uncertainty in a

$$\frac{s_y}{y} = x \left(\frac{s_a}{a} \right)$$

Let's say: $a = 4.00 \pm 0.01$
 $x = 3$

$$y = 4.00^3 = 64.0$$

$$\frac{s_y}{y} = 3 \times \left(\frac{0.01}{4.00} \right) = 0.075$$

To find the absolute SD: $s_y = y (= 64.0) \times (\pm 0.075) = \pm 4.8$

$$64.0 \pm 4.8$$

This rule applies only when the exponent x has negligible uncertainty.

See example 4-5

Standard Deviations of Logarithms and Antilogarithms

Consider: $y = \log a$

The absolute standard deviation of the logarithm of a number is determined by the relative standard deviation of the number.

$$s_y = 0.434 \frac{s_a}{a}$$

and for $y = \text{antilog } a = 10^a$

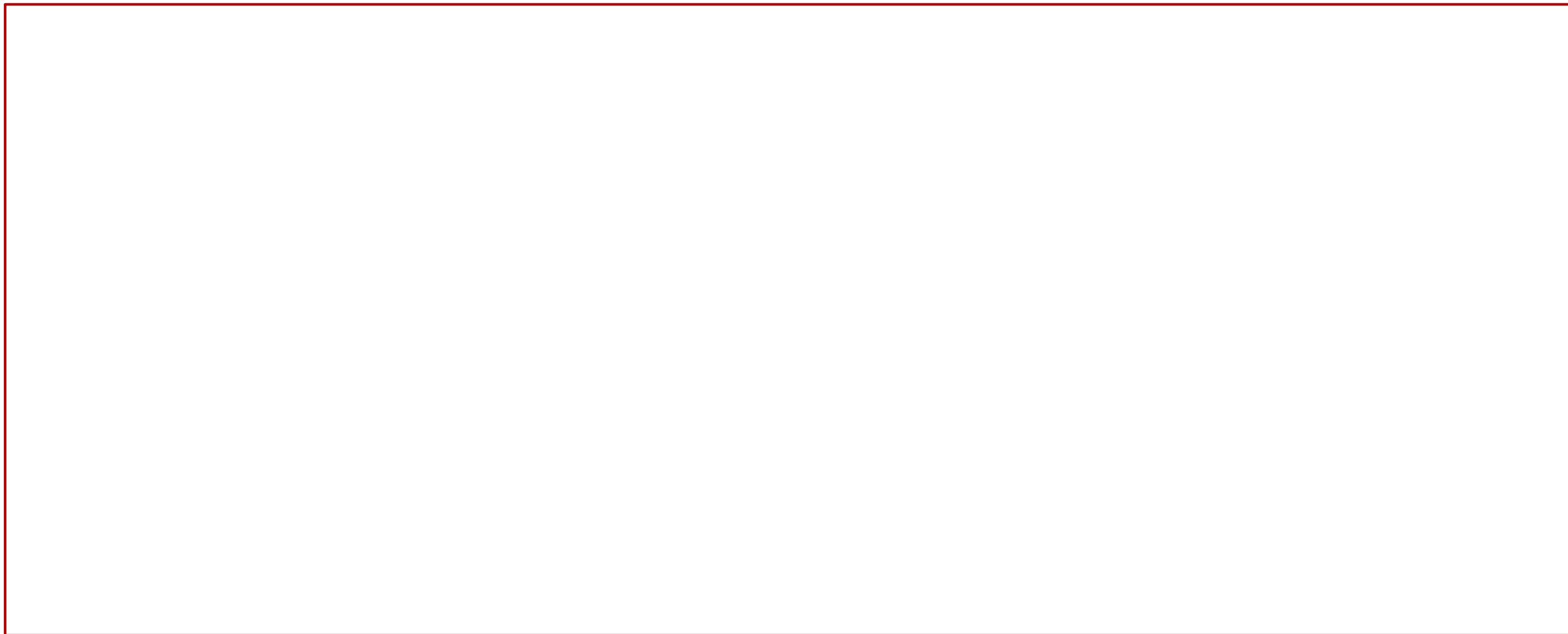
Conversely, the relative standard deviation of the antilogarithm of a number is determined by the absolute standard deviation of the number.

$$\frac{s_y}{y} = 2.303 s_a$$

Practice Problem

What is the pH?

$$[H^+] = (4.45 \pm 0.03) \times 10^{-5} M$$



Propagation of Error: Random vs. Systematic Error

Vast majority of measured phenomena exhibit random variation

Watch out for exceptions: Some quantities (e.g., atomic masses, calibrated glassware) have tabulated uncertainties or known systematic bias.

Propagation of Random Uncertainty

Calibrated Volumetric Pipet

Experimental Calibration:

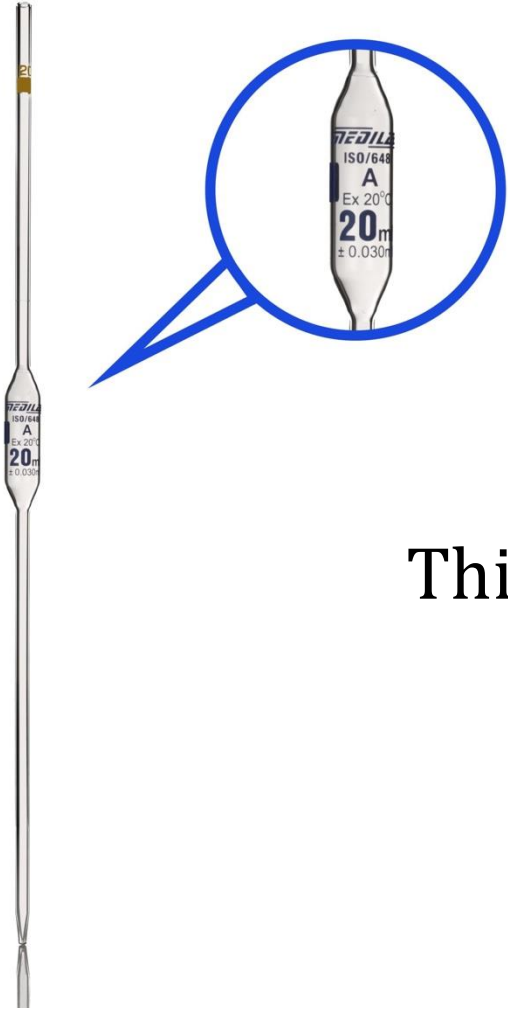
$$24.991 \pm 0.006 \text{ mL}$$

This is a *random error* and combines quadratically:

$$99.964 \pm \sqrt{0.006^2 + 0.006^2 + 0.006^2 + 0.006^2} \text{ mL}$$

$$99.964 \pm 0.012 \text{ mL}$$

$$99.96 \pm 0.01 \text{ mL}$$



Propagation of Systematic Error

Uncalibrated Volumetric Pipet

$$25.00 \pm 0.03 \text{ mL}$$

Because the *error is systematic*, it adds directly (not quadratically).

Used 4 times to deliver 100 mL:

$$(25.00 \pm 0.03 \text{ mL}) \times 4 = 100.0 \pm 0.12 \text{ mL}$$

Propagation of Uncertainty with Systematic Error

Errors in atomic masses are not random.

The errors are determined by the distribution of isotopes in nature.

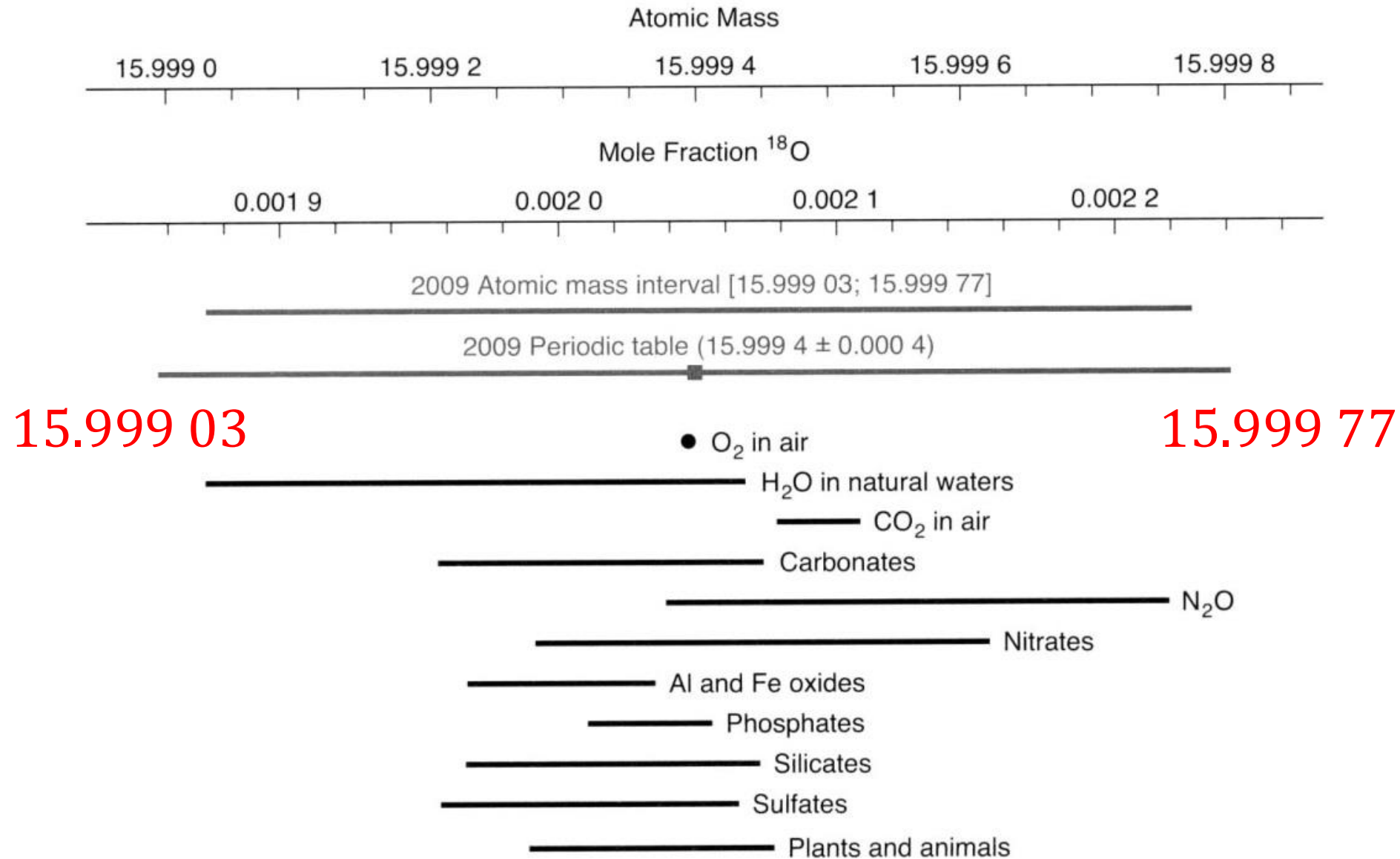
Thus the uncertainty distribution is not normal but approximately rectangular.

Oxygen: has 3 stable isotopes, 16, 17 and 18, with 16 dominating at 99.76%.

$15.999 \pm 0.001 \text{ g/mol}$

						10 2 He helium 4.0026 ± 0.0001

Propagation of Uncertainty with Systematic Error



Propagation of Uncertainty with Systematic Error

O₂

$$O = 15.9994 \pm 0.0004 \text{ g/mol}$$

 **Tabulated Uncertainty**

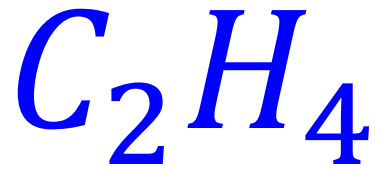
Range for O₂=

$$2 \times 15.9998 - 2 \times 15.9990 = \\ 31.9996 - 31.9980 \text{ g/mol}$$

$$31.9988 \pm 0.0008 \text{ g/mol}$$

The uncertainty limits add linearly because they are systematic, not random.

Propagation of Uncertainty with Systematic Error



First, find the standard uncertainties

$$C = 12.0106 \pm 0.0010$$

$$H = 1.00798 \pm 0.00014$$

$$2C: \quad 2(12.0106 \pm 0.0010) = 24.0212 \pm \mathbf{0.0020} \leftarrow 2 \times 0.0010$$

$$4H: \quad 4(1.00798 \pm 0.00014) = \underline{4.03192 \pm \mathbf{0.00056}} \leftarrow 4 \times 0.00014$$
$$28.05312 \pm ?$$

$$28.05312 \pm \sqrt{0.0020^2 + 0.00056^2} = 28.0531 \pm 0.0021 \text{ g/mol}$$

$$\mathbf{28.053 \pm 0.0021 \text{ g/mol}}$$